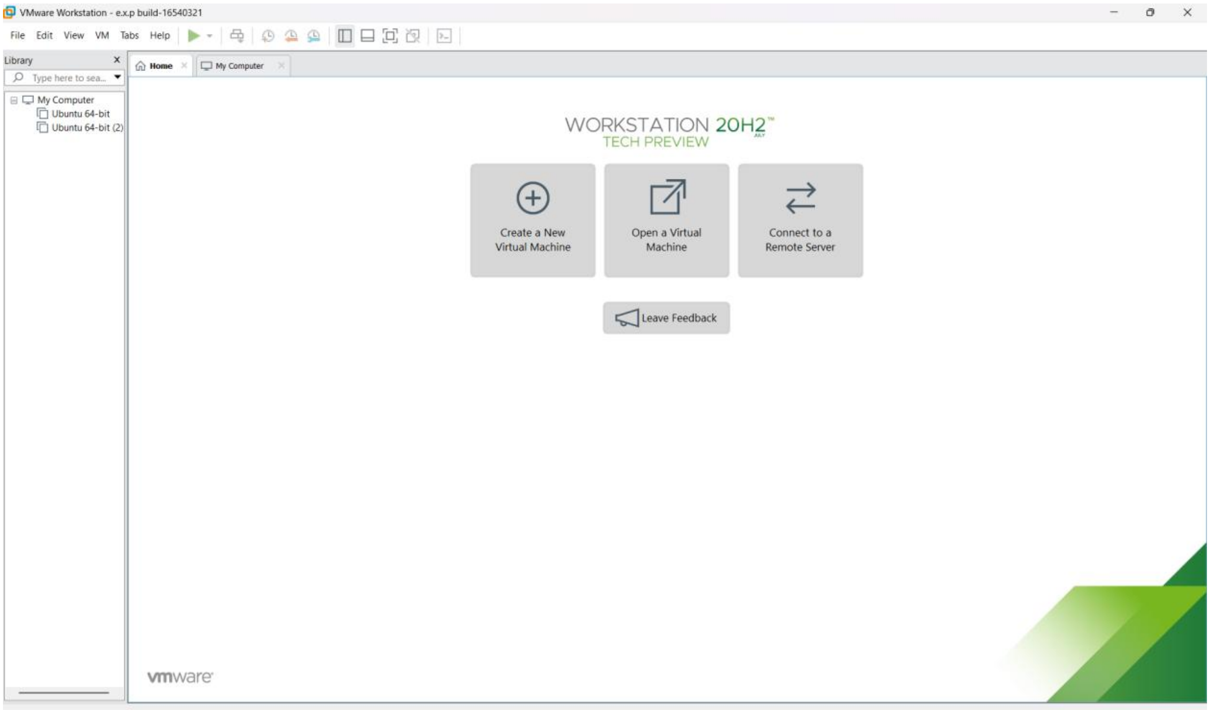
Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux).

AIM:
To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows /linux).

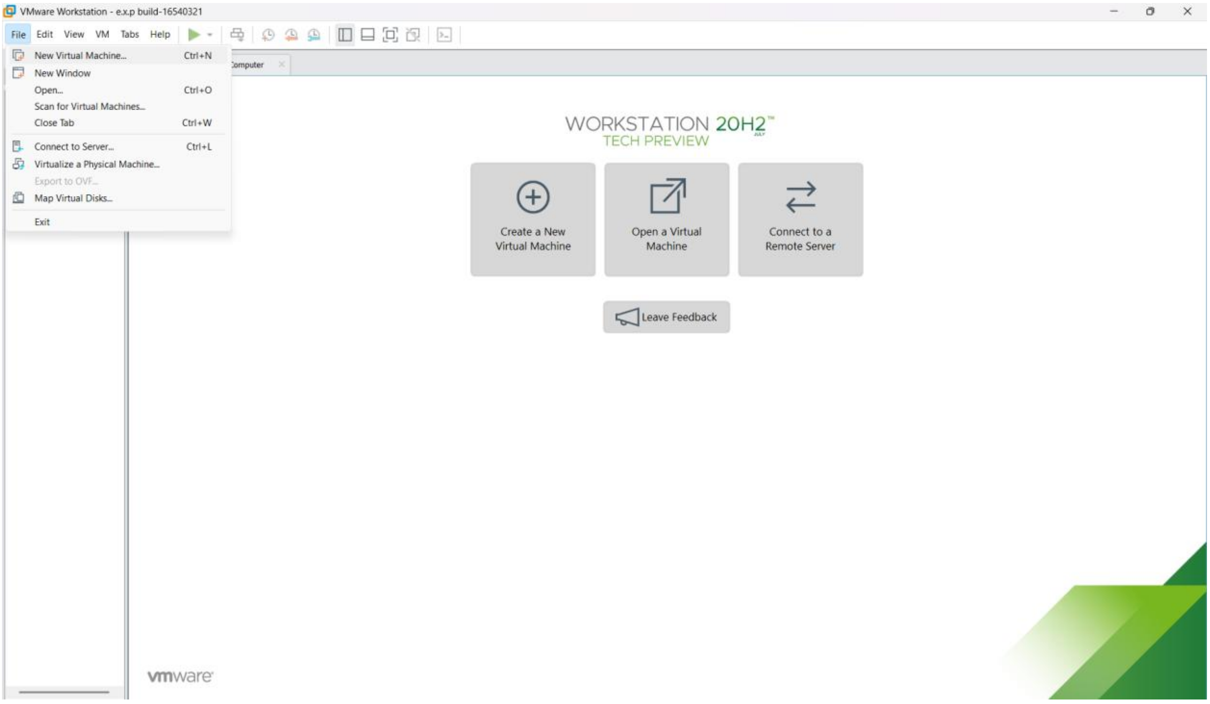
PROCEDURE:
STEP 1: Dowload VMware workstation and installed as type 2hypervisor.
STEP2: Dowload ubuntu or tiny OS as iso image file.
STEP 3: In VMware workstation->create new VM.
STEP 4: Do the basic configuration settings.
STEP 5: Created tiny OS virtual machine.
STEP 6: Launch the VM.

IMPLEMENTATION:

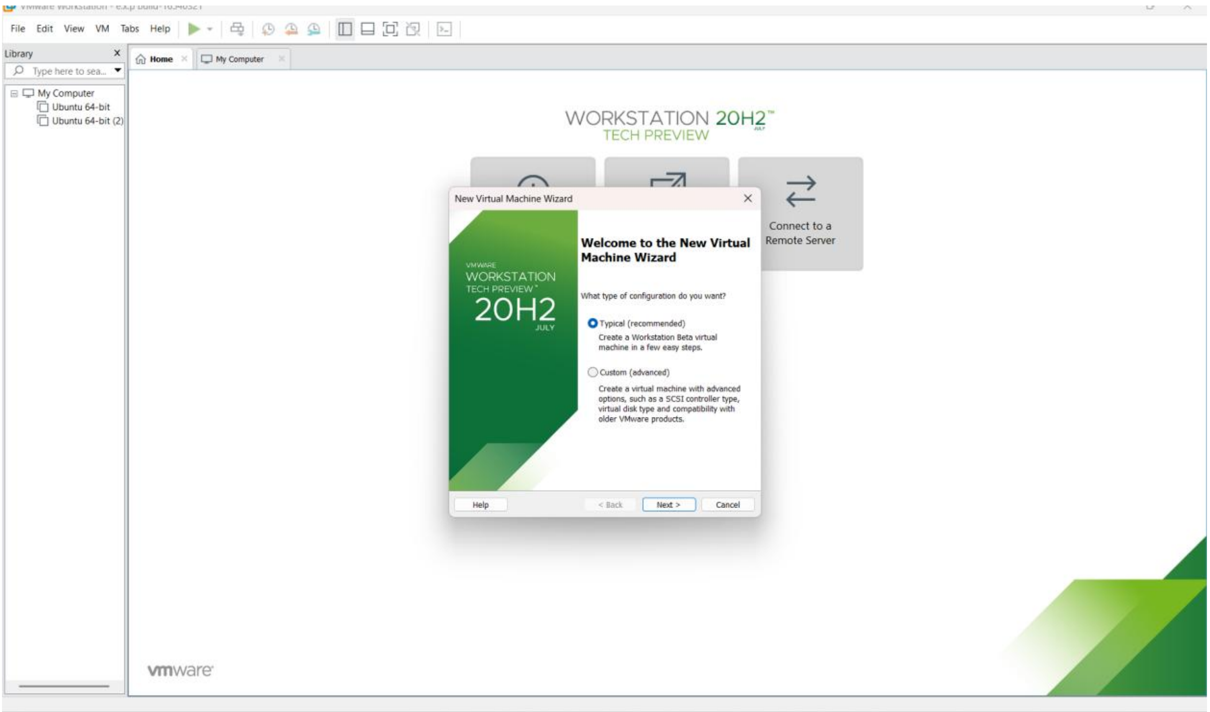
STEP 1:DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

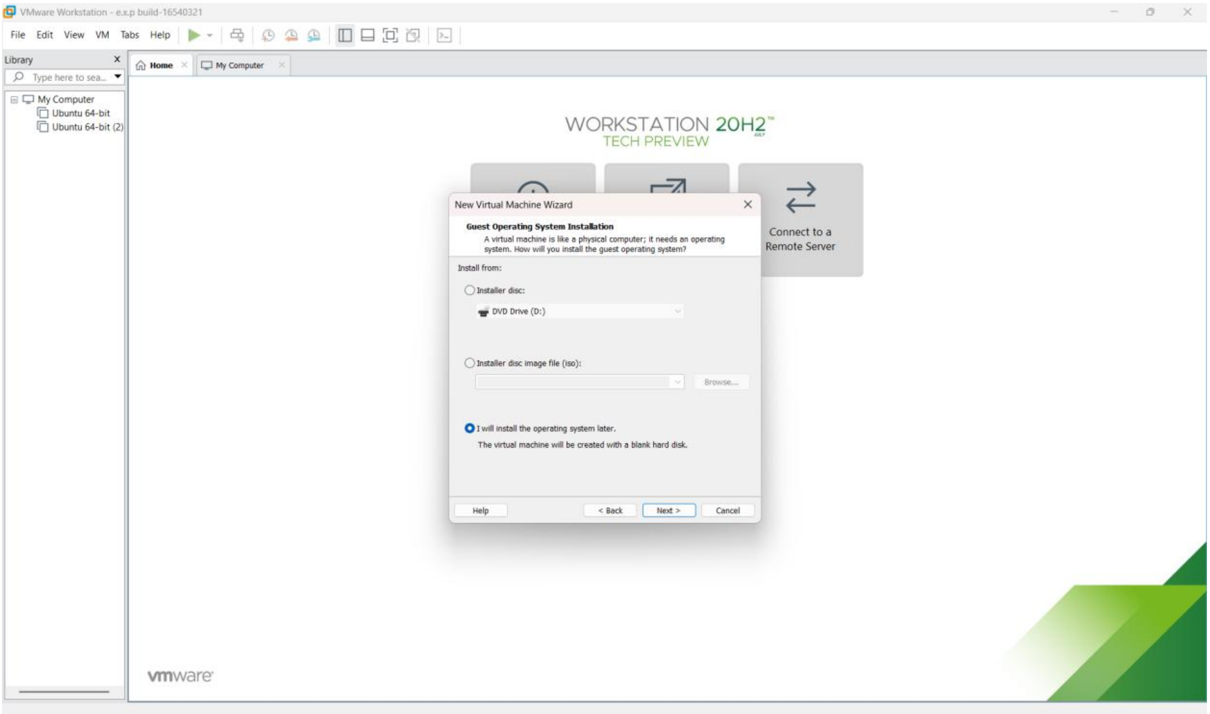


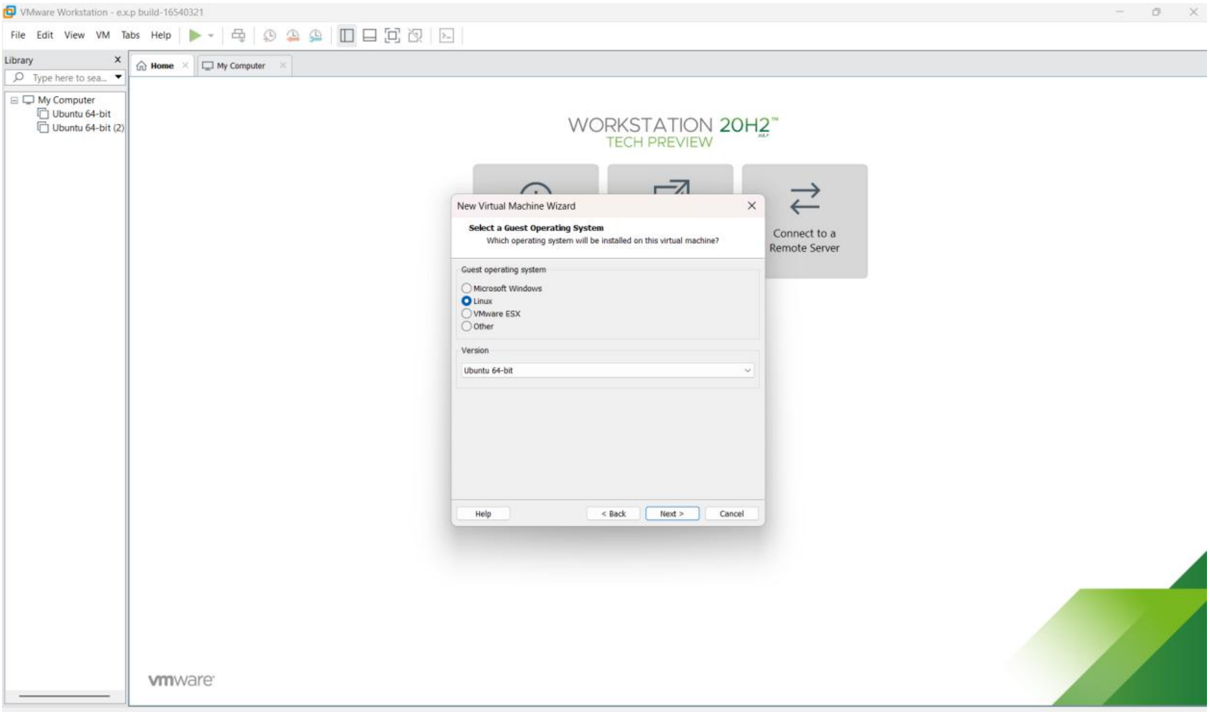
STEP 2: IN VMWARE WORKSTATION->CREATE NEW VM

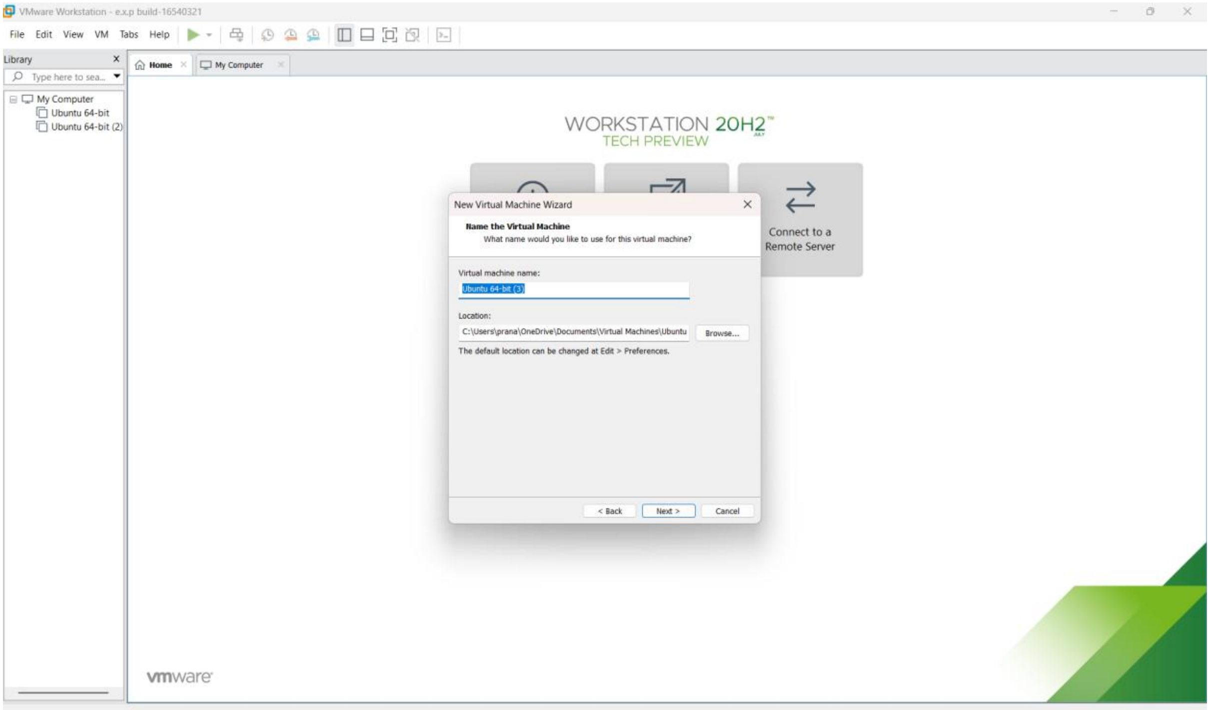


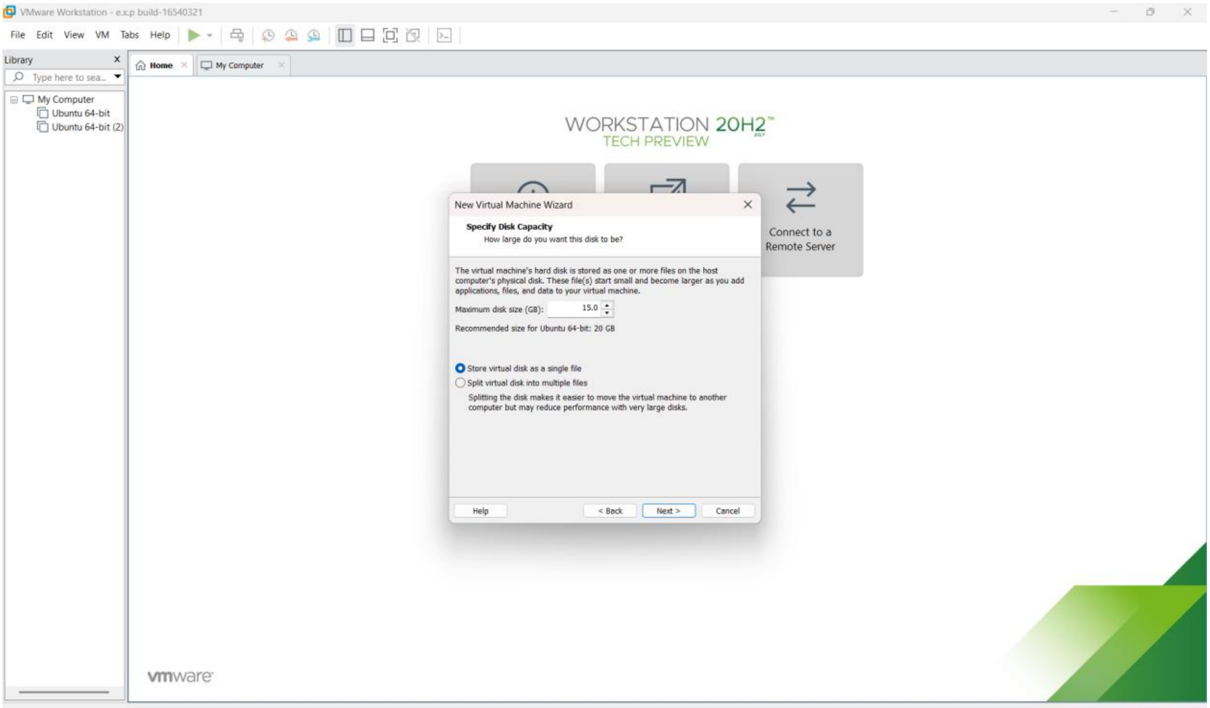
STEP 3: DO THE BASIC CONFIGURATION SETTINGS.

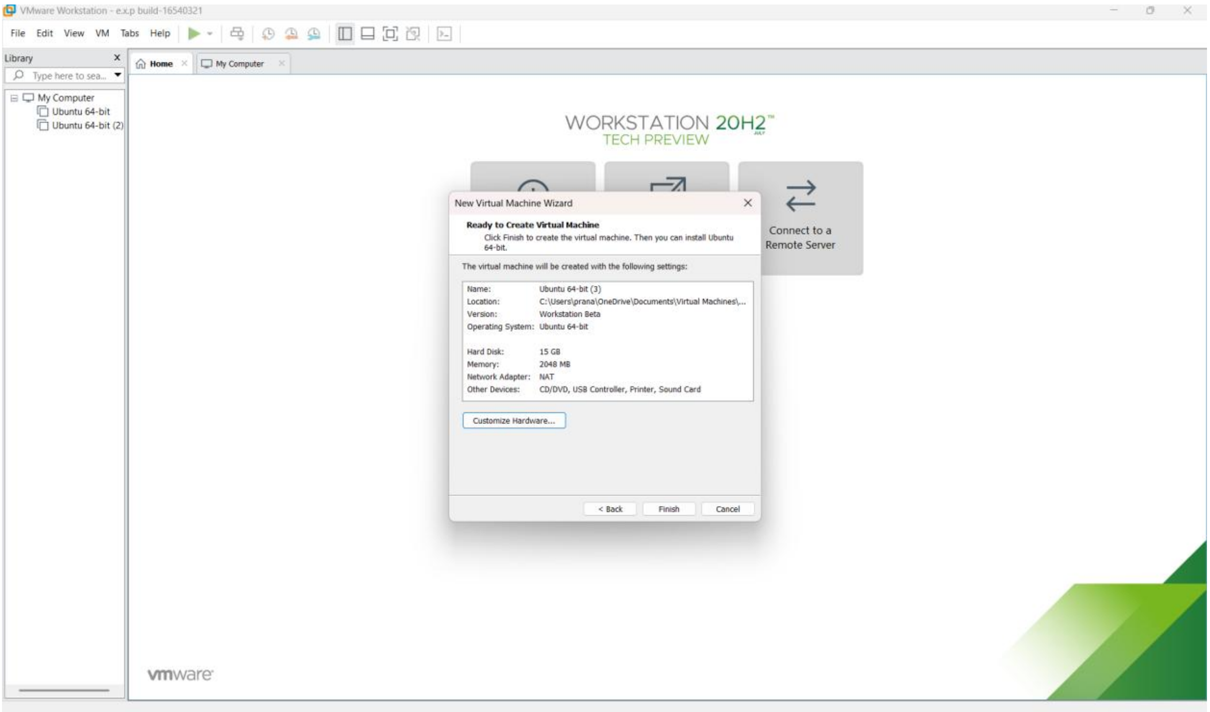



VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Home
My Computer
WORKSTATION 20H2
TECH PREVIEW
Connect to a Remote Server
New Virtual Machine Wizard
Guest Operating System Installation
A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?
Install from:
Installer disc:
DVD Drive (D:)
Installer disc image file (iso):
Browse...
I will install the operating system later.
The virtual machine will be created with a blank hard disk.
Help
< Back
Next >
Cancel
vmware

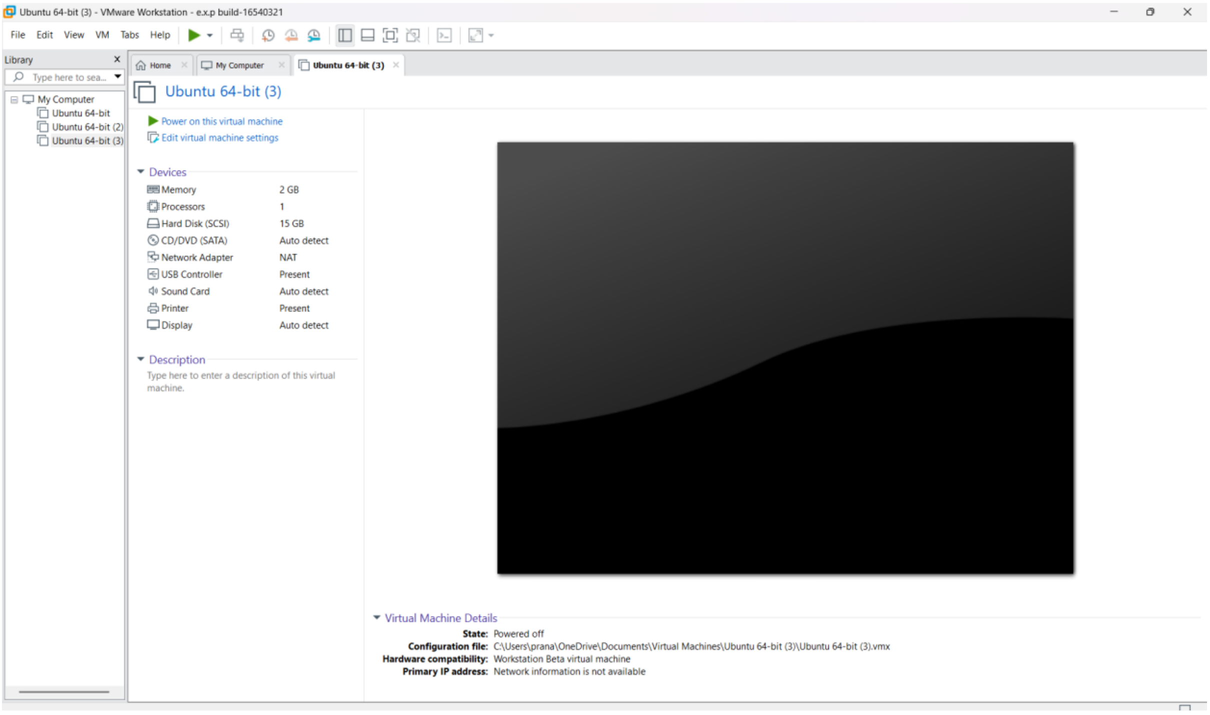

VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Home
My Computer
WORKSTATION 20H2
TECH PREVIEW
Connect to a Remote Server
New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Ubuntu 64-bit
Help
< Back
Next >
Cancel
vmware

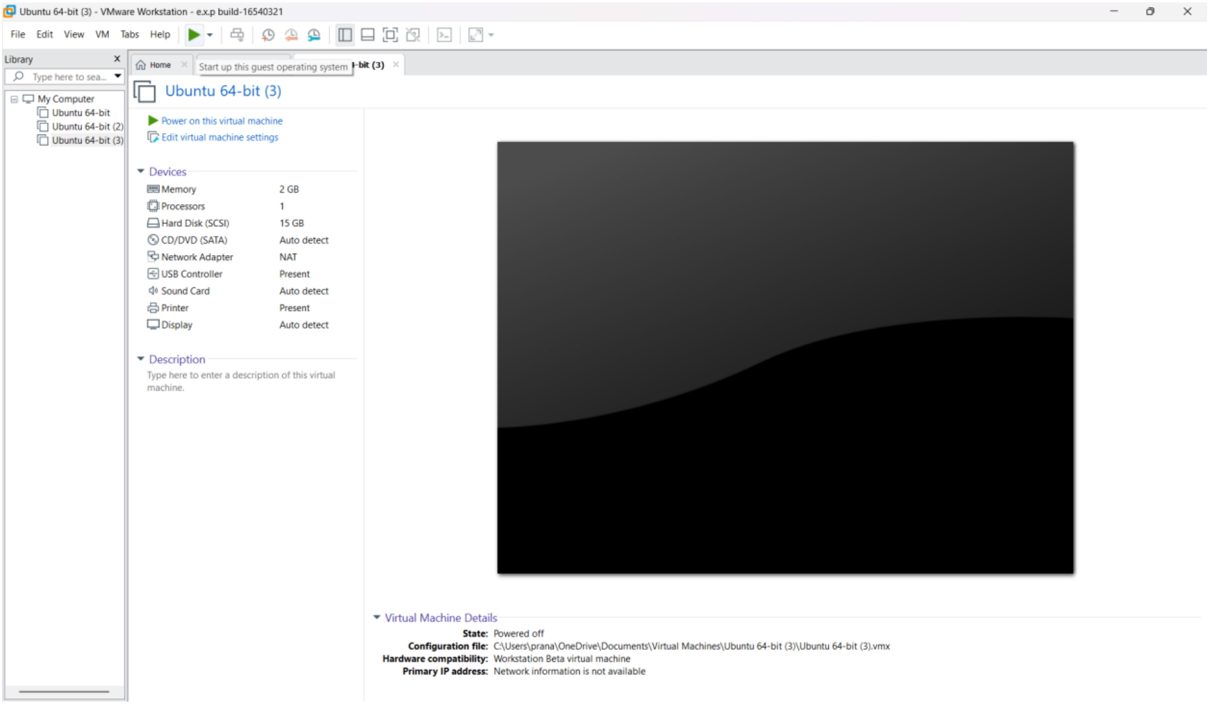

VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Home
My Computer
WORKSTATION 20H2
TECH PREVIEW
Connect to a Remote Server
New Virtual Machine Wizard
Name the Virtual Machine
What name would you like to use for this virtual machine?
Virtual machine name:
Ubuntu 64-bit (3)
Location:
C:\Users\prana\OneDrive\Documents\Virtual Machines\Ubuntu
Browse...
The default location can be changed at Edit > Preferences.
< Back
Next >
Cancel
vmware


VMware Workstation - e.x.p build-16540321
File Edit View VM Tabs Help
Library
Type here to sea...
My Computer
Ubuntu 64-bit
Ubuntu 64-bit (2)
Home
My Computer
WORKSTATION 20H2
TECH PREVIEW
Connect to a Remote Server
New Virtual Machine Wizard
Specify Disk Capacity
How large do you want this disk to be?
The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.
Maximum disk size (GB): 15.0
Recommended size for Ubuntu 64-bit: 20 GB
Store virtual disk as a single file
Split virtual disk into multiple files
Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.
Help
< Back
Next >
Cancel
vmware

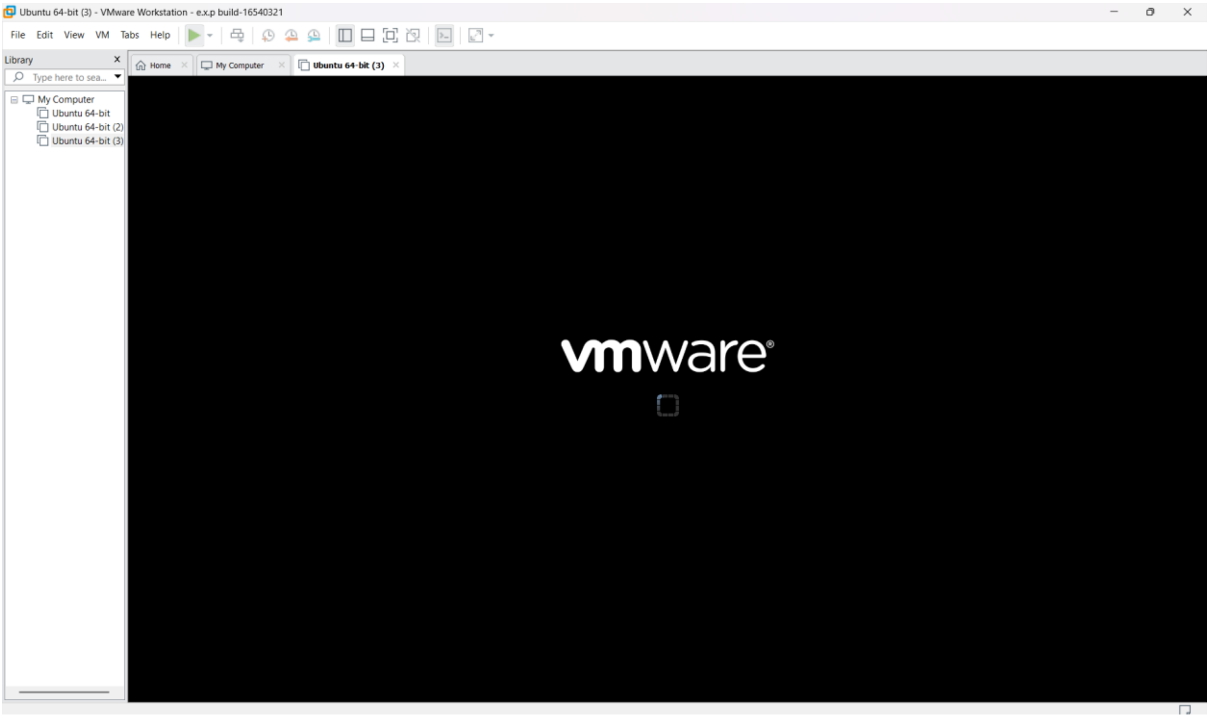


STEP 4: CREATED TINYOS VIRTUAL MACHINE





STEP 5: LAUNCH THE VM



Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim:

To create and configure a virtual machine with 1 CPU, 2 GB RAM, and 15 GB storage using VirtualBox as a Type-2 virtualization software.

Procedure:

Step 1: Open Oracle VirtualBox on the host computer.

Step 2: Select the already created Ubuntu VM and open Settings. **Step 3:** Go to System → Motherboard and set the memory to 2048 MB (2 GB).

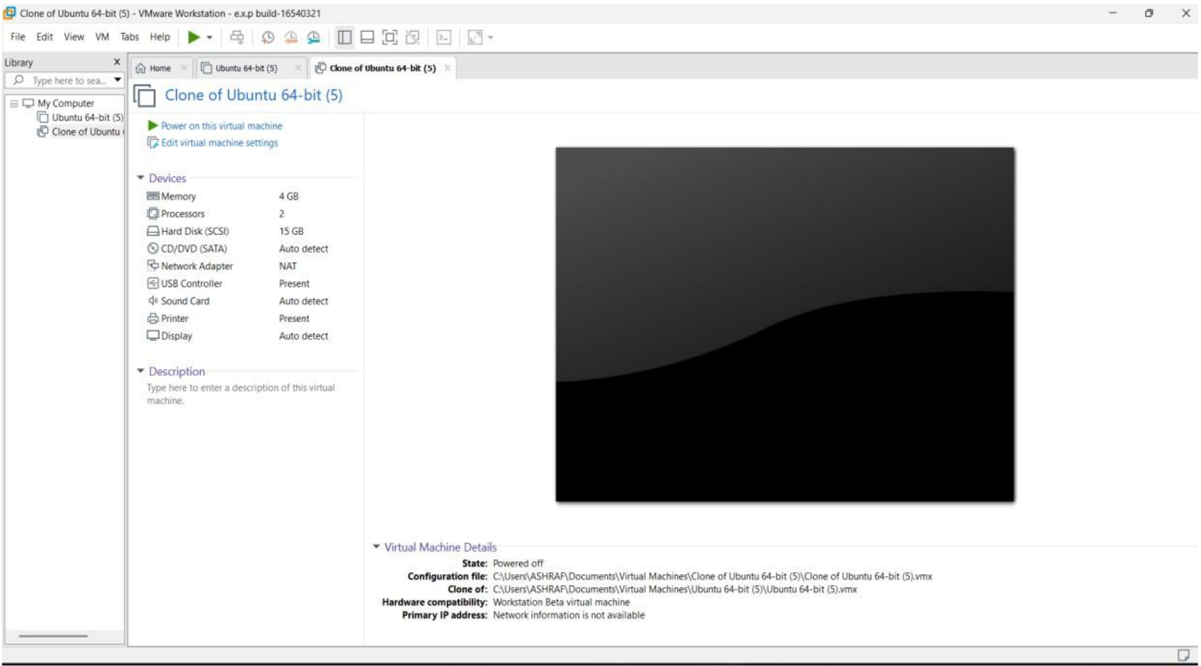
Step 4: Go to System → Processor and set the processor count to 1 CPU.

Step 5: Open Storage and verify that the virtual hard disk is configured with 15 GB storage.

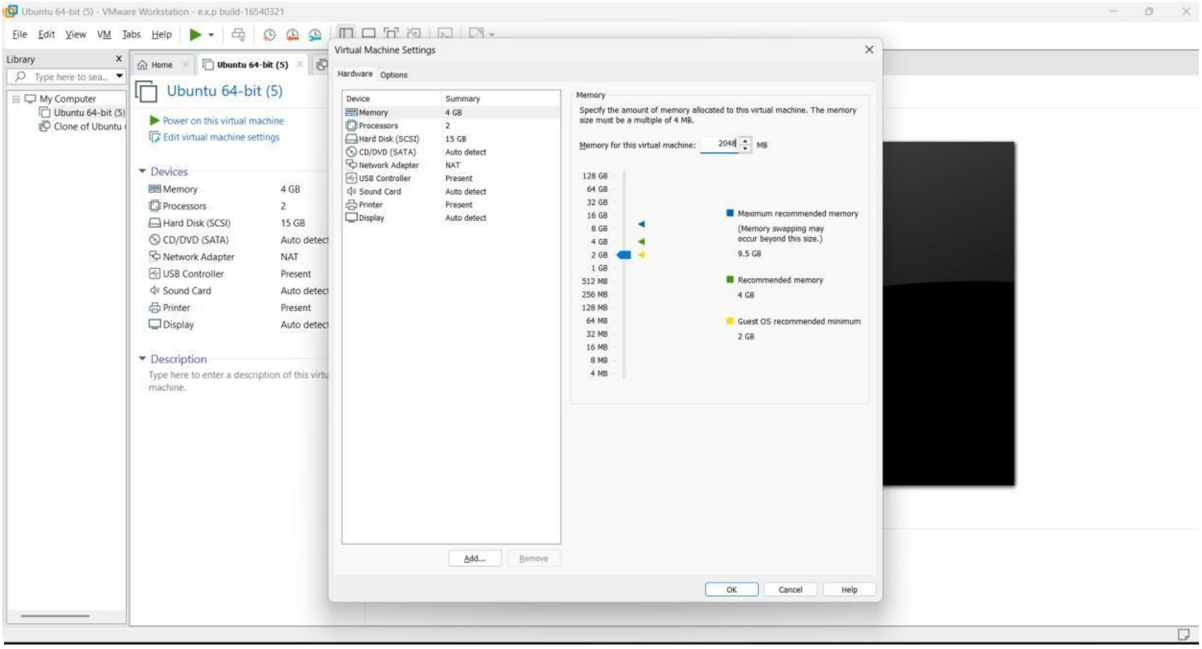
Step 6: Click OK, start the Ubuntu VM, and verify that it boots successfully.

IMPLEMENTATION:

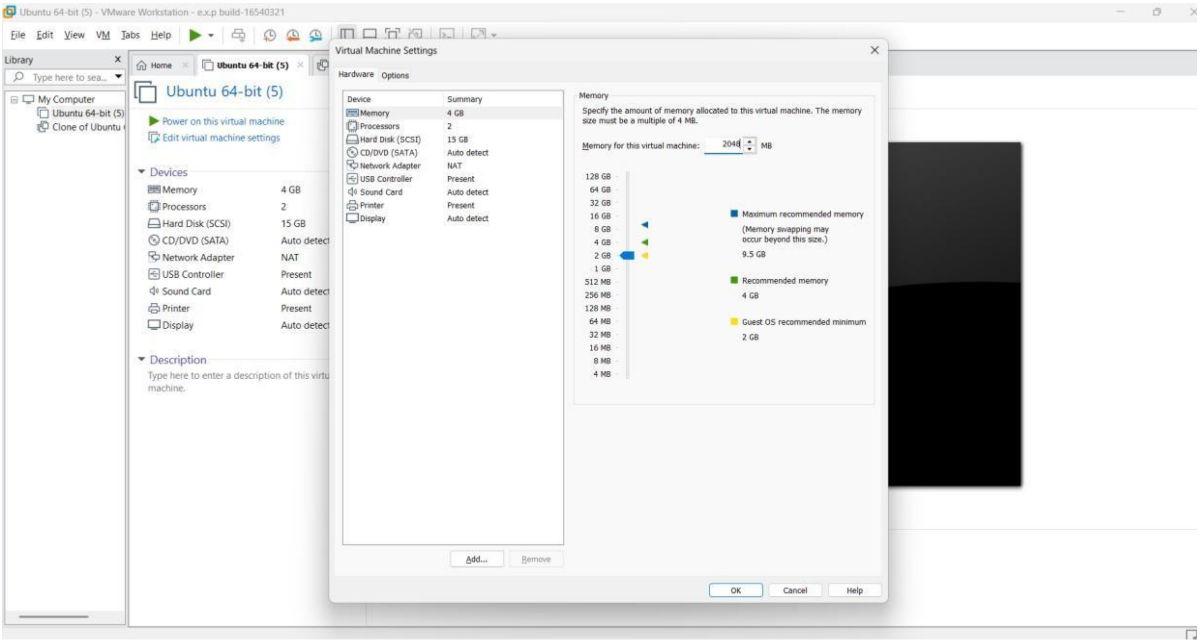
Step 1: Open Oracle VirtualBox on the host computer.



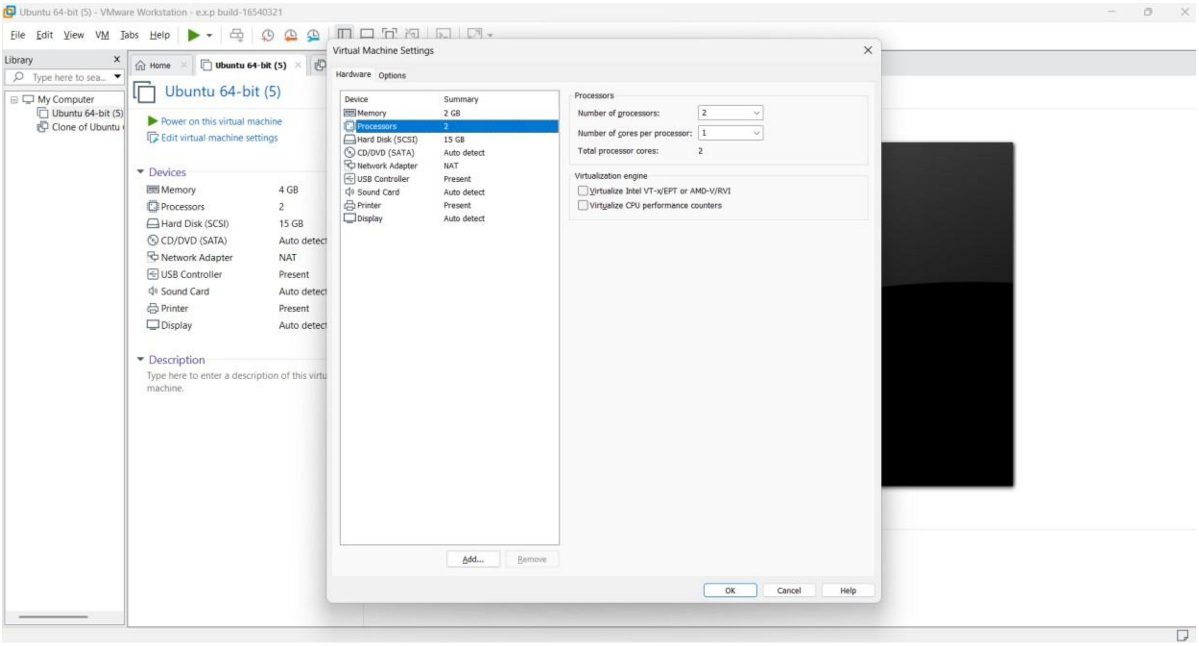
Step 2: Select the already created Ubuntu VM and open Settings.



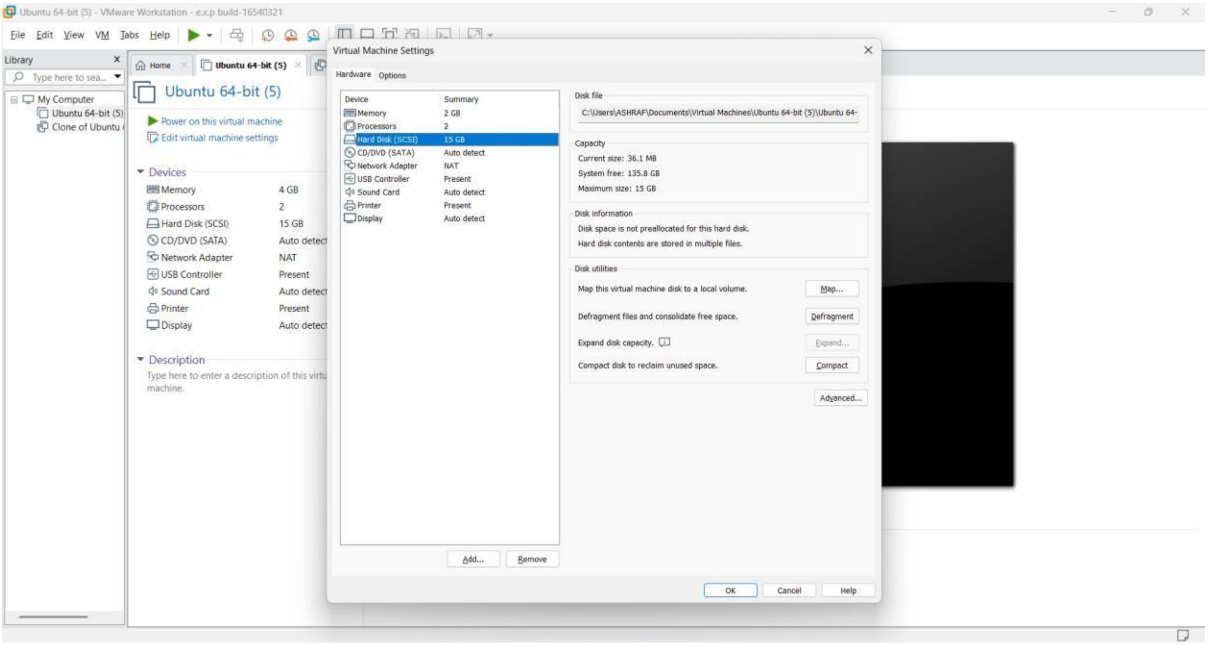
Step 3: Go to System → Motherboard and set the memory to 2048 MB (2 GB).



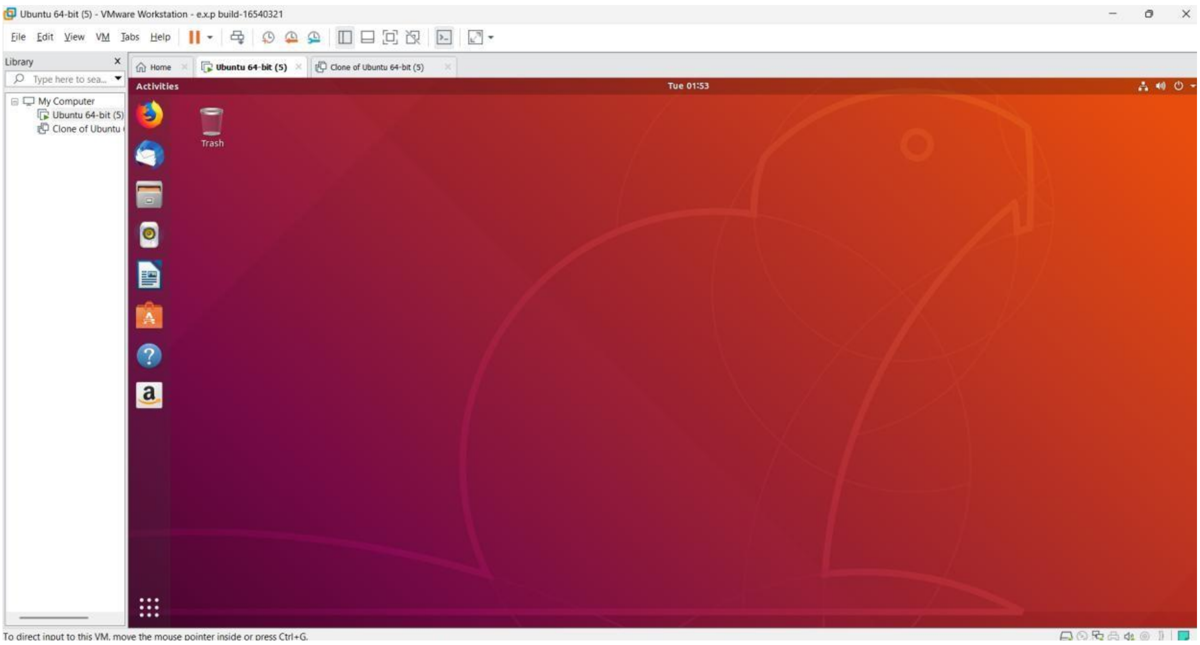
Step 4: Go to System → Processor and set the processor count to 1 CPU.



Step 5: Open Storage and verify that the virtual hard disk is configured with 15 GB storage.



Step 6: Click OK, start the Ubuntu VM, and verify that it boots successfully.



Create a Virtual Hard Disk and allocate the storage using VM ware Workstation

Aim:

To create a virtual hard disk and allocate storage to it using VirtualBox.

Procedure:

Step 1: Open VMware Workstation and select the existing Ubuntu virtual machine.

Step 2: Click Edit virtual machine settings.

Step 3: Select Hard Disk and click Add to create a new virtual hard disk.

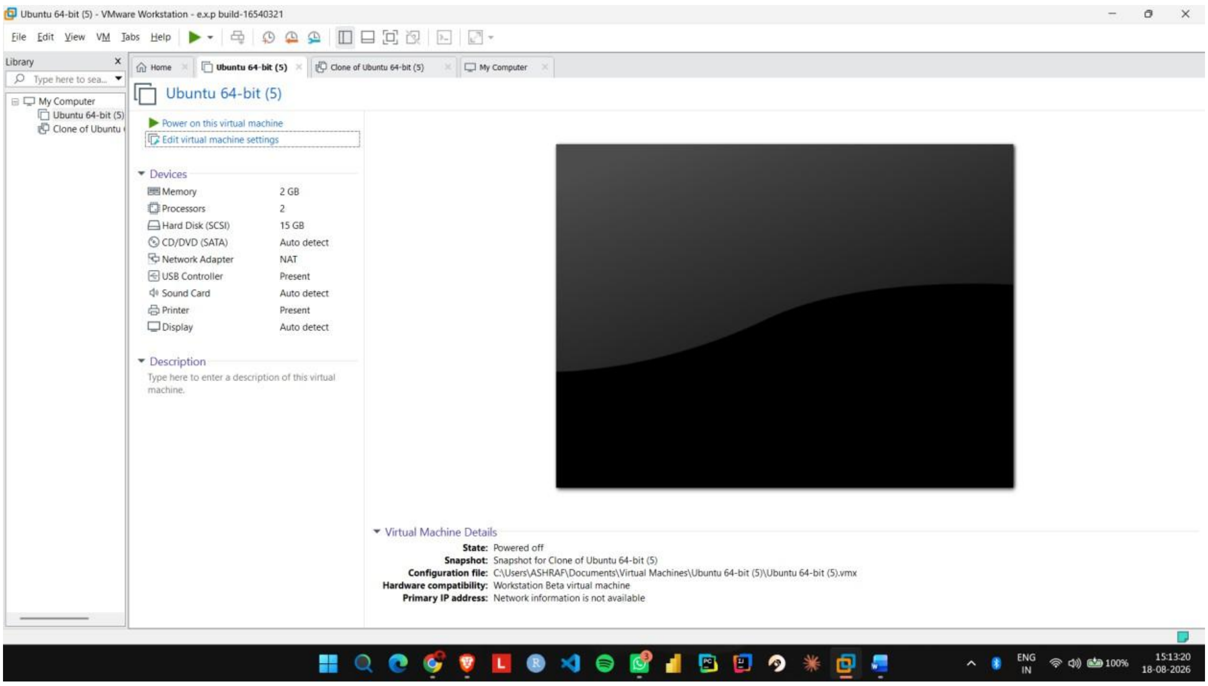
Step 4: Select Hard Disk → SCSI as the virtual disk type.

Step 5: Select Create a new virtual disk and allocate the required storage, such as 15 GB.

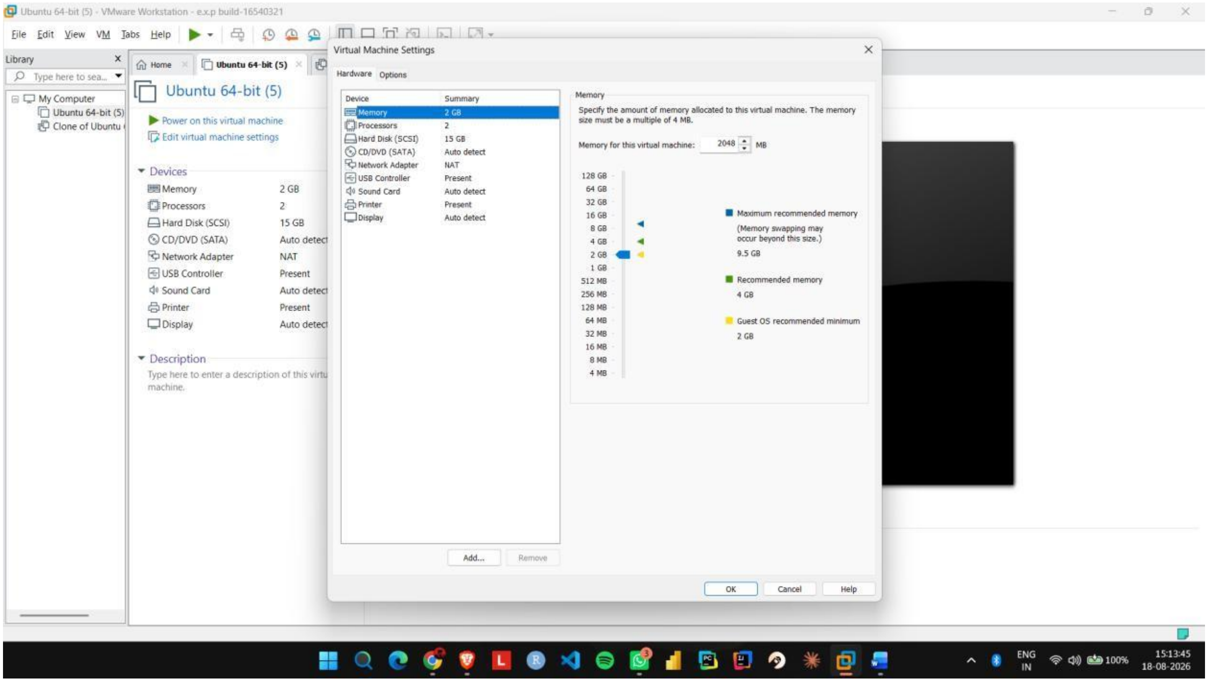
Step 6: Click Finish and verify that the new virtual hard disk is attached to the Ubuntu VM.

IMPLEMENTATION:

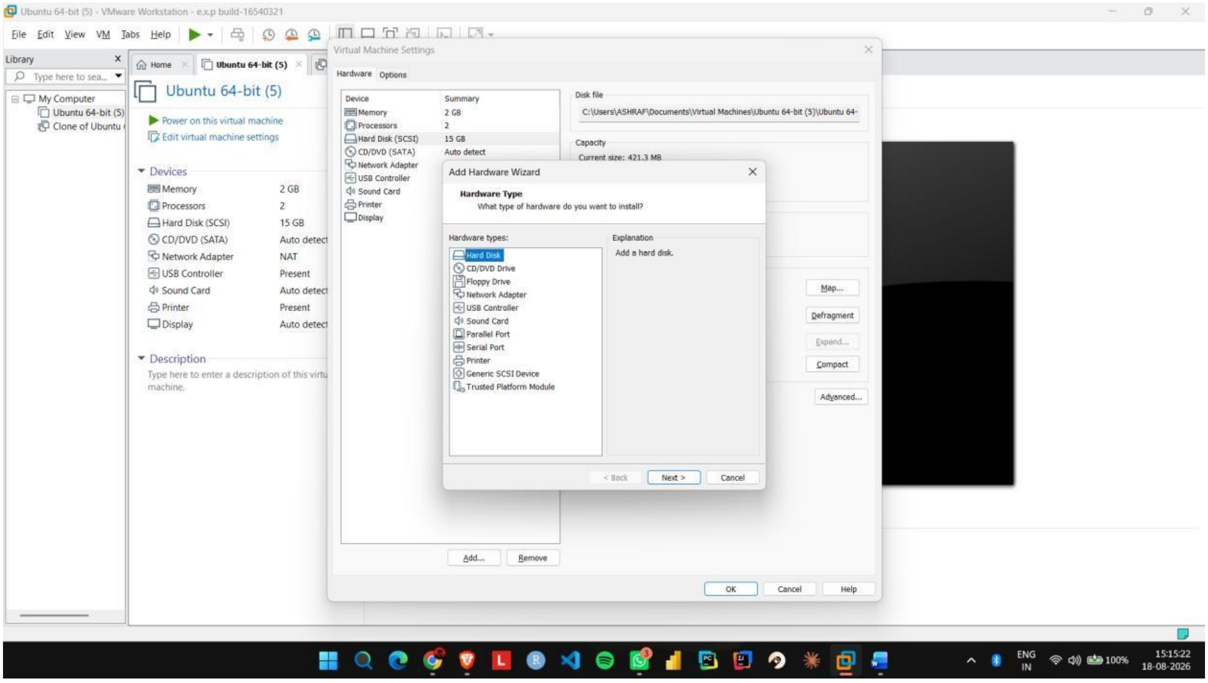
Step 1: Open VMware Workstation and select the existing Ubuntu virtual machine.



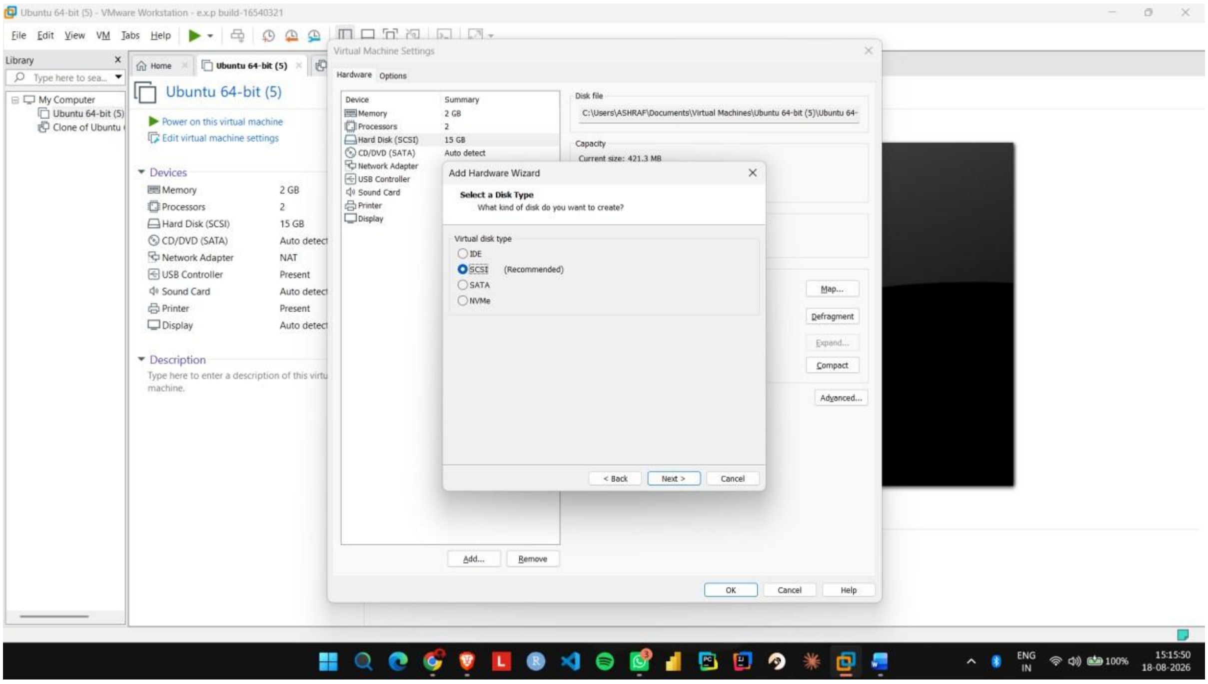
Step 2: Click Edit virtual machine settings.



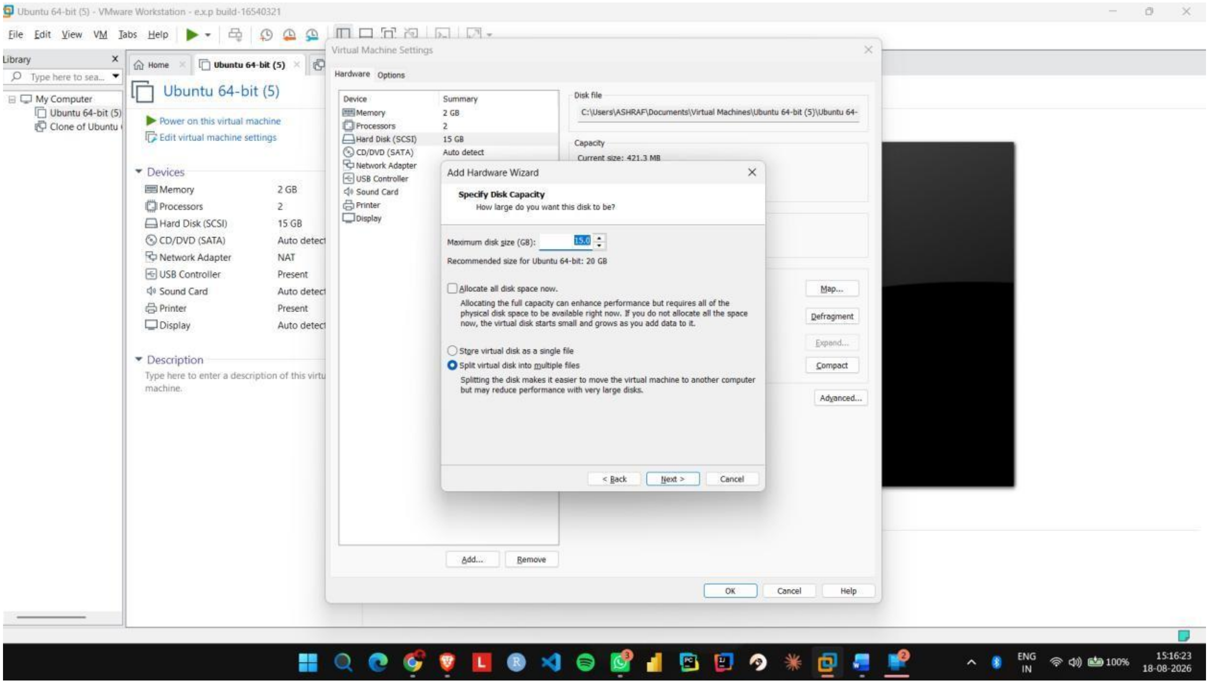
Step 3: Select Hard Disk and click Add to create a new virtual hard disk.



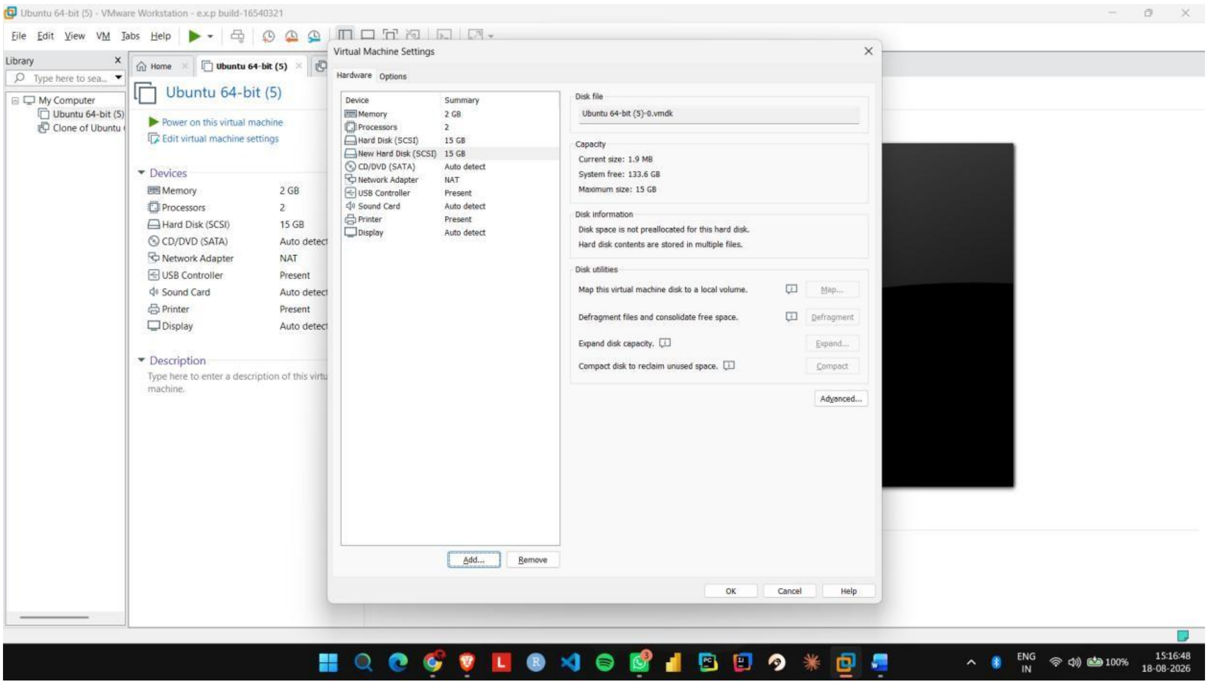
Step 4: Select Hard Disk → SCSI as the virtual disk type.



Step 5: Select Create a new virtual disk and allocate the required storage, such as 15 GB.



Step 6: Click Finish and verify that the new virtual hard disk is attached to the Ubuntu VM.



Create a Snapshot of a VM and Test it by loading the Previous Version/Cloned VM

Aim:

To create a snapshot of a virtual machine and restore the previous version using VMware Workstation.

Procedure:

Step 1: Start the Ubuntu VM in VMware Workstation and create a test file or folder.

Step 2: Shut down or suspend the VM and open the Snapshot option.

Step 3: Select Take Snapshot and enter a name such as Before_Changes.

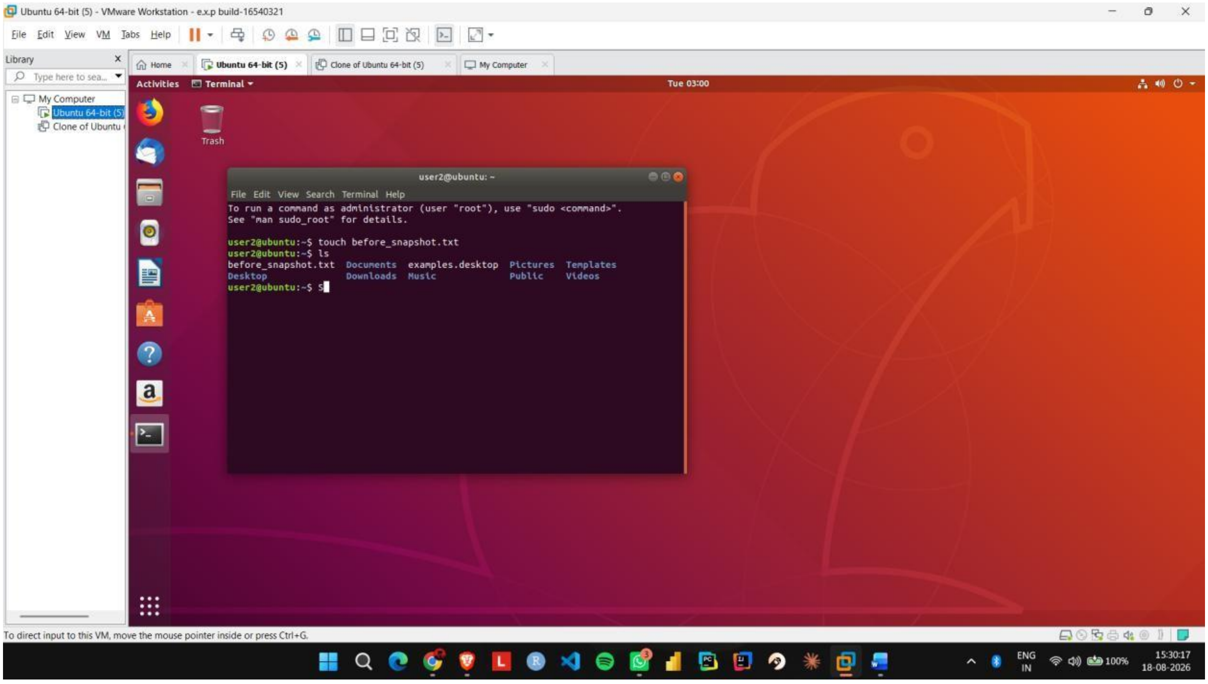
Step 4: Start the VM again and make some changes, such as creating a new file.

Step 5: Open Snapshot Manager, select the Before_Changes snapshot, and click Go To/Restore.

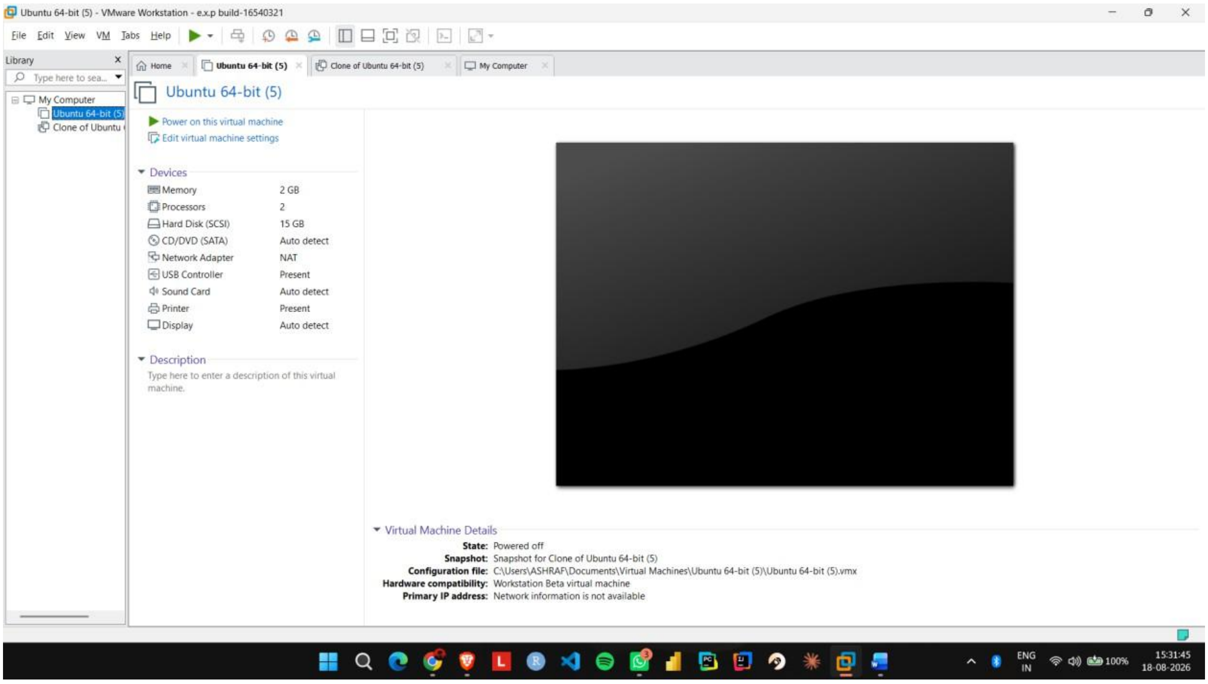
Step 6: Start the VM and verify that the changes made after the snapshot have been removed.

IMPLEMENTATION:

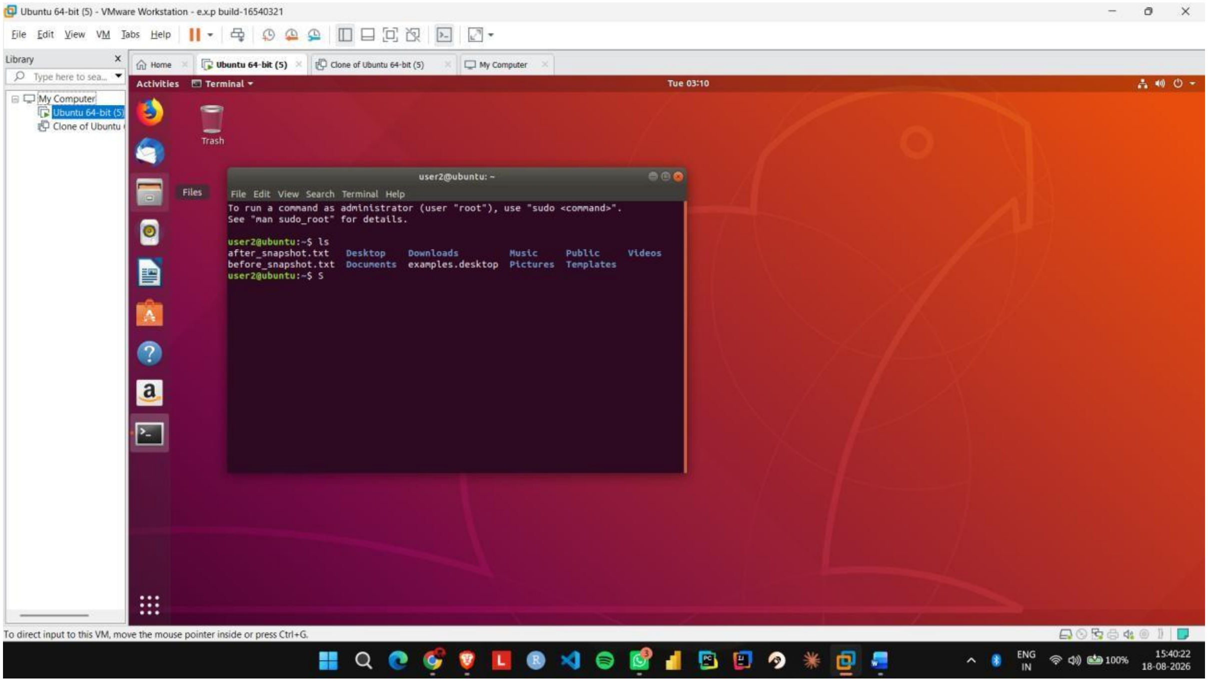
Step 1: Start the Ubuntu VM in VMware Workstation and create a test file or folder.



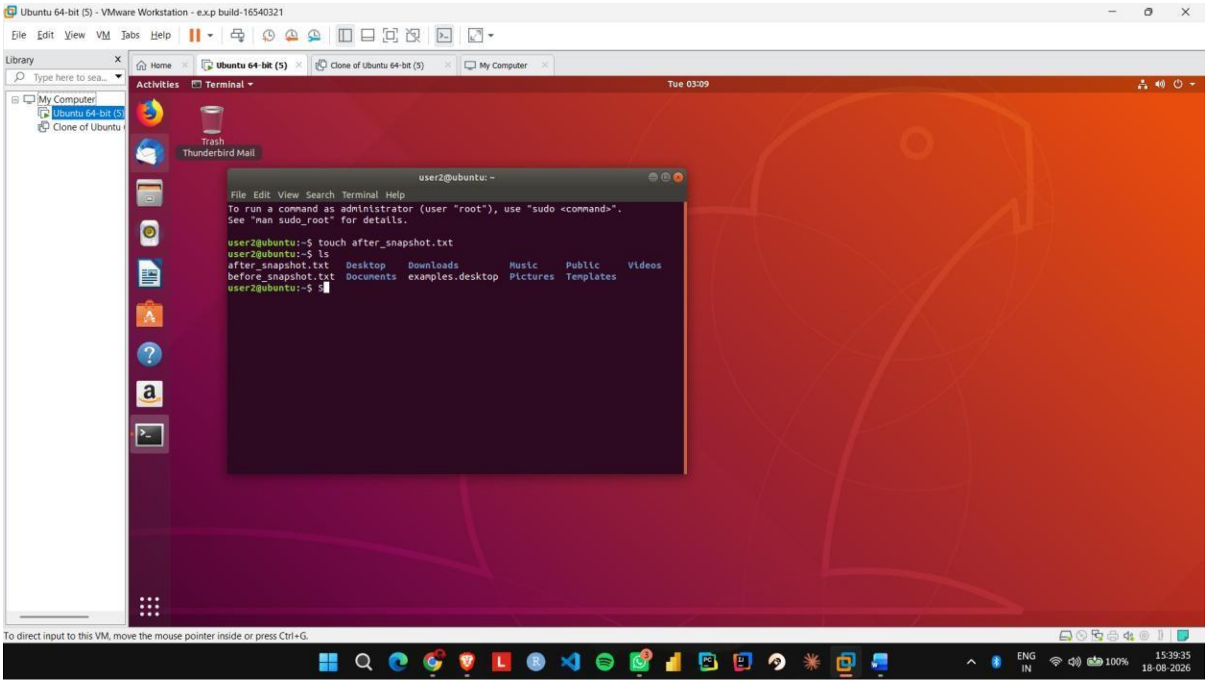
Step 2: Shut down or suspend the VM and open the Snapshot option.



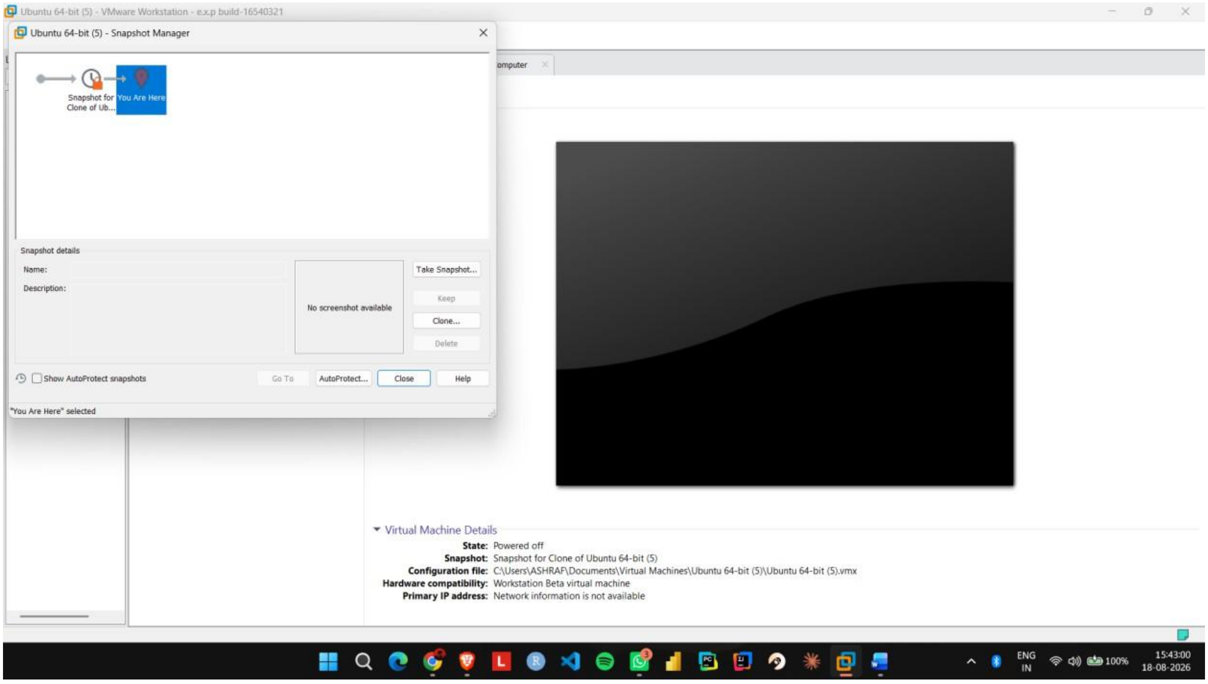
Step 3: Select Take Snapshot and enter a name such as Before Changes.



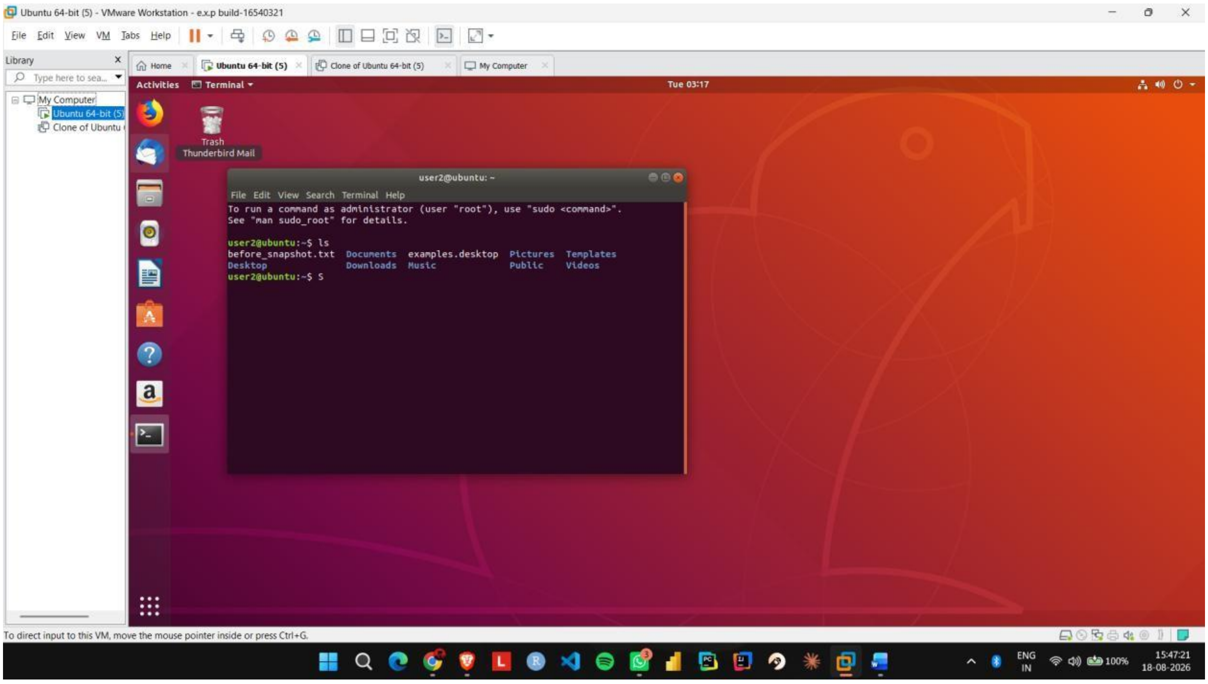
Step 4: Start the VM again and make some changes, such as creating a new file.



Step 5: Open Snapshot Manager, select the Before_Changes snapshot, and click Go To/Restore.



Step 6: Start the VM and verify that the changes made after the snapshot have been removed.



Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim:

To create a clone of an existing virtual machine and test the cloned virtual machine using VMware Workstation.

Procedure:

Step 1: Shut down the existing Ubuntu VM completely.

Step 2: Open VMware Workstation and select VM → Manage → Clone.

Step 3: Select the current state of the virtual machine as the cloning source.

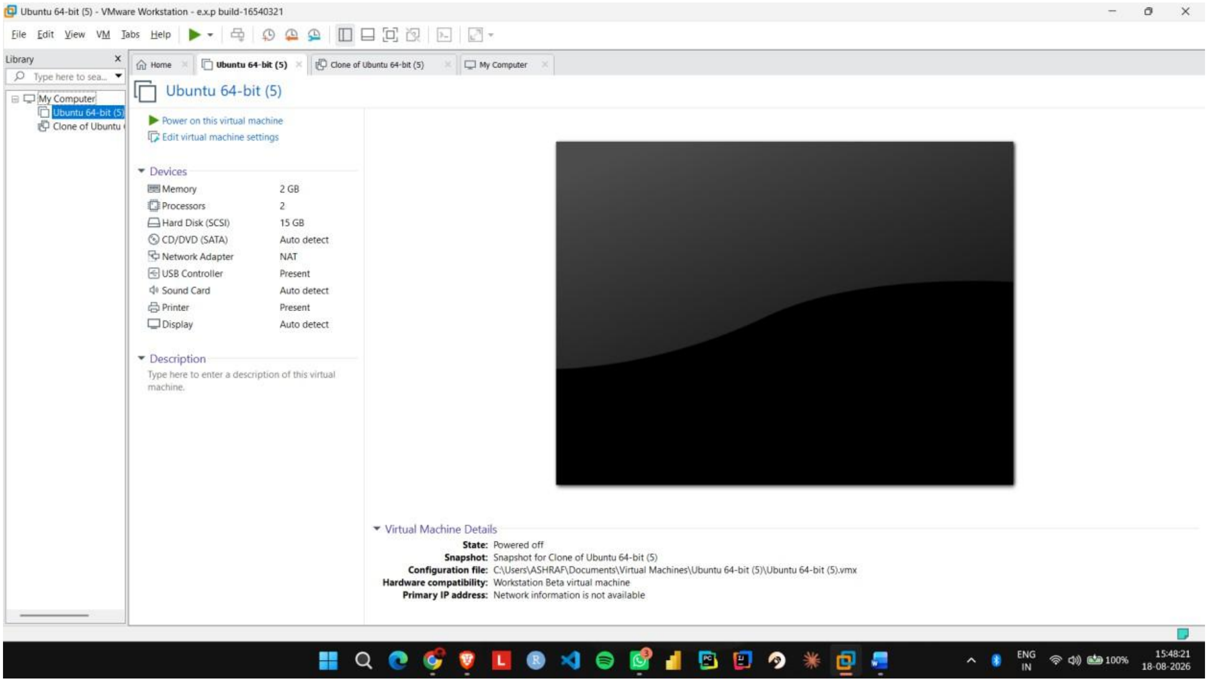
Step 4: Select Create a Full Clone and give the cloned VM a name such as Ubuntu_Clone.

Step 5: Complete the cloning process and start the Ubuntu_Clone
virtual machine.

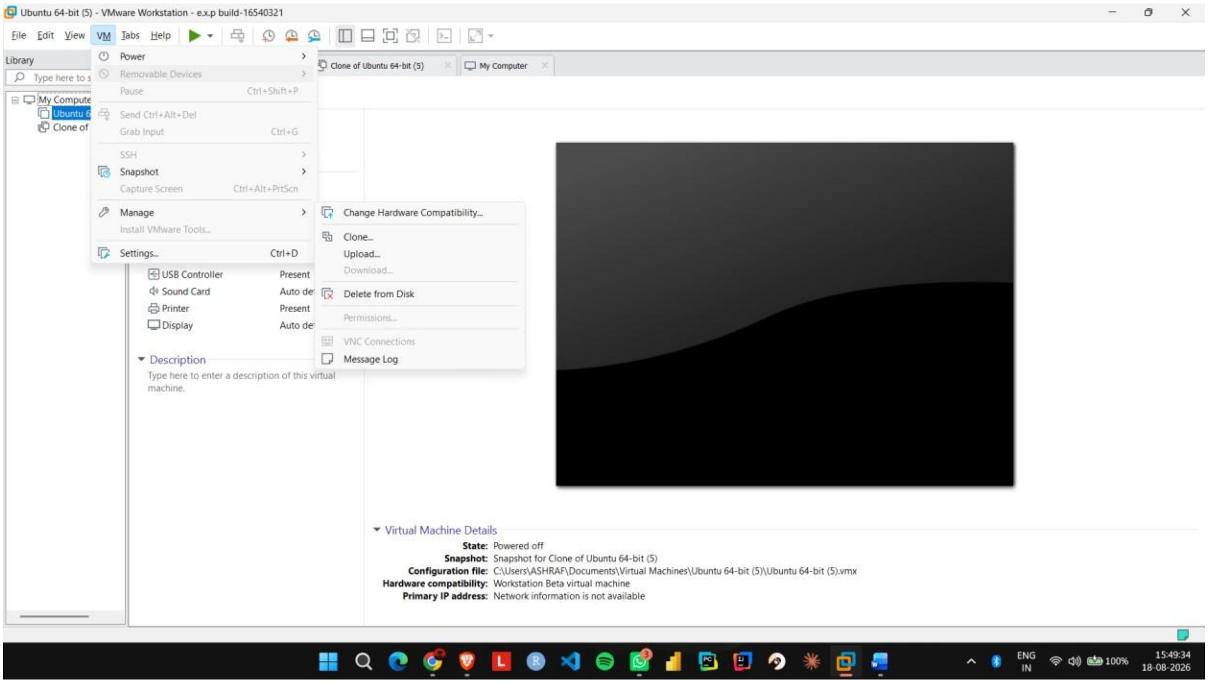
Step 6: Create a test file in the cloned VM and verify that it does not appear in the original Ubuntu VM.

IMPLEMENTATION:

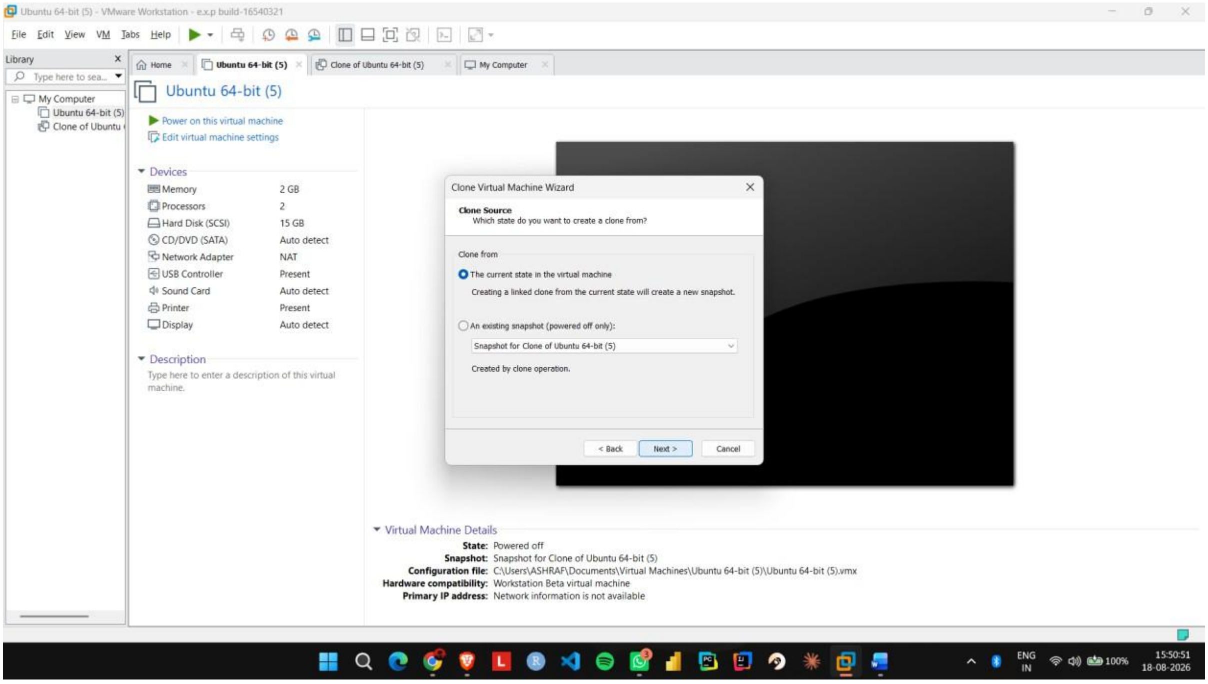
Step 1: Shut down the existing Ubuntu VM completely.



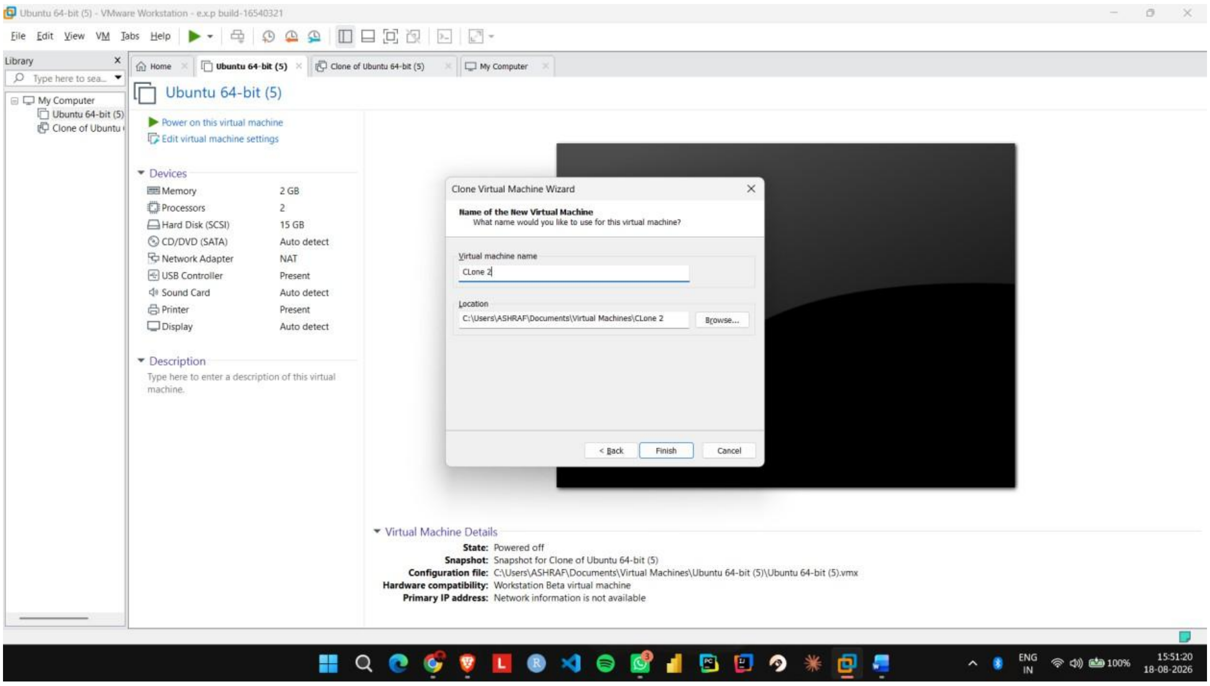
Step 2: Open VMware Workstation and select VM → Manage → Clone.



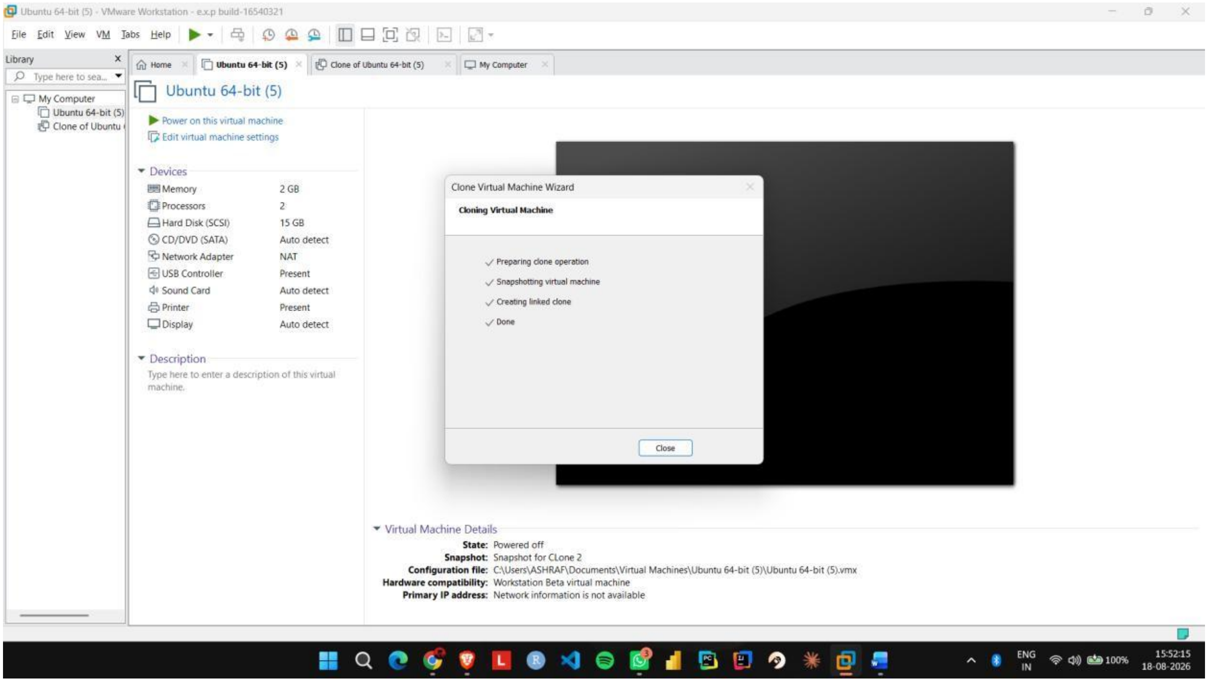
Step 3: Select the current state of the virtual machine as the cloning source.



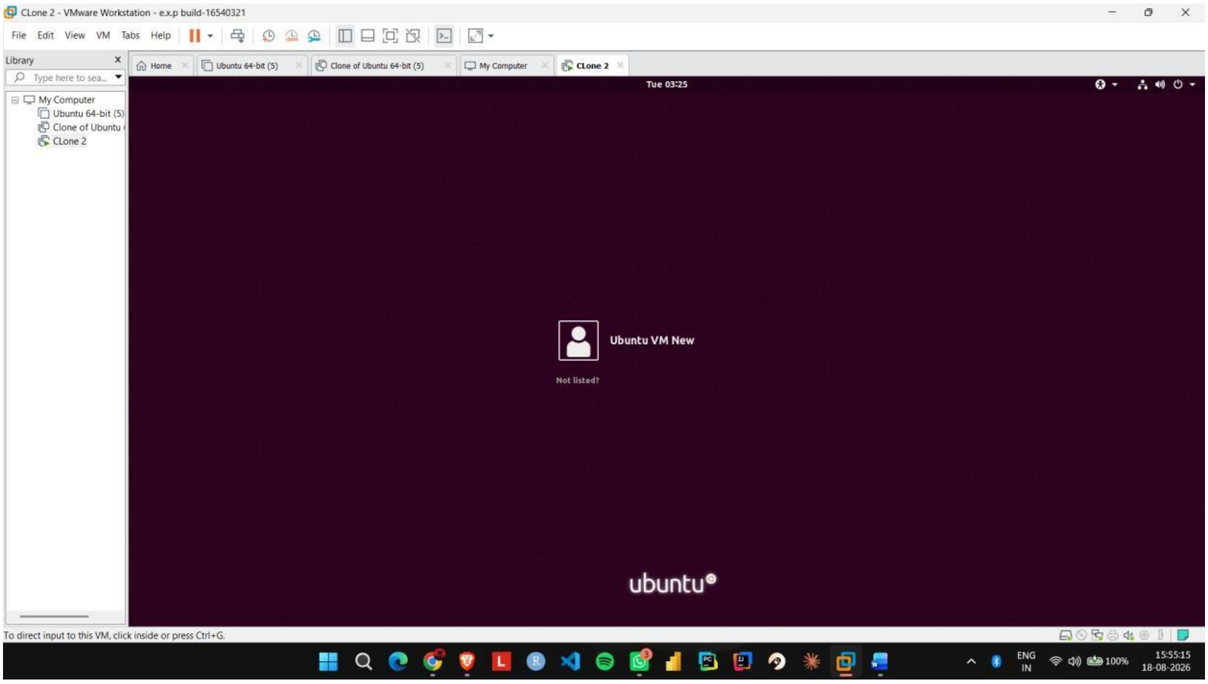
Step 4: Select Create a Full Clone and give the cloned VM a name such as Ubuntu_Clone.



Step 5: Complete the cloning process and start the Ubuntu_Clone virtual machine.



Step 6: Create a test file in the cloned VM and verify that it does not appear in the original Ubuntu VM.



Change Hardware compatibility of a VM (Either by clone/create new one) which is already created and configured and run simple unix commands

Aim:

To change the hardware configuration and compatibility of an existing virtual machine using VMware Workstation.

Procedure:

Step 1: Shut down the existing Ubuntu VM and open Edit virtual machine settings.

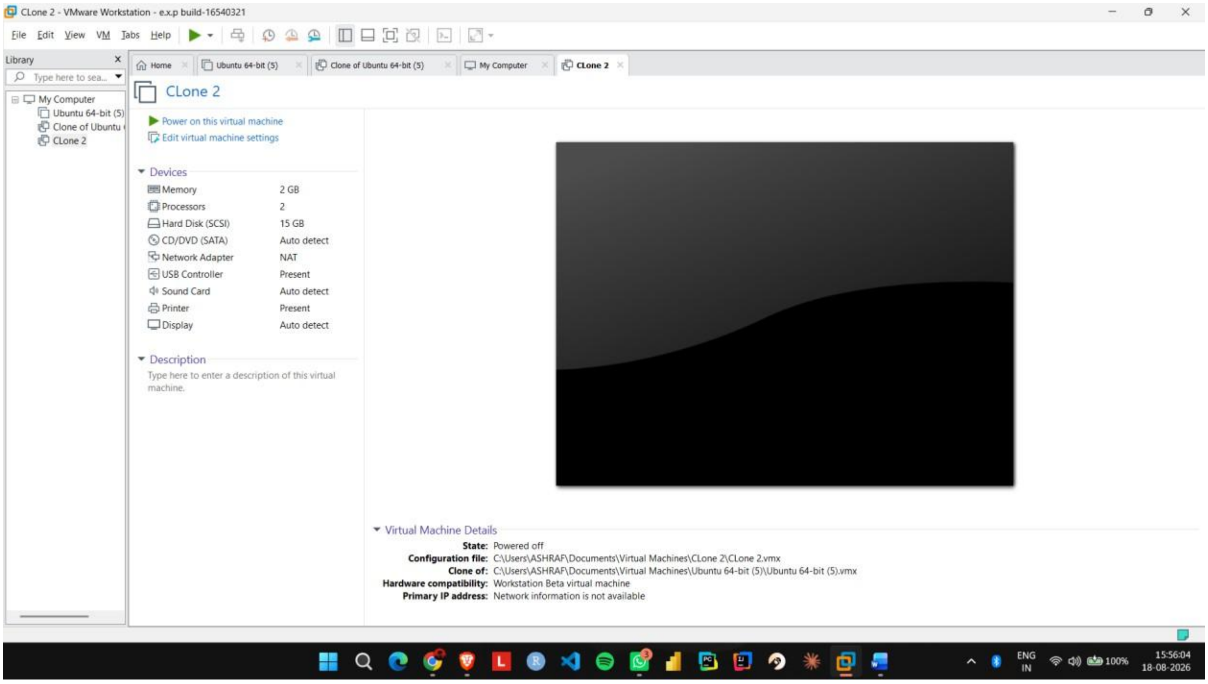
Step 2: Go to Hardware and change the memory from 2 GB to 4 GB. **Step 3:** Change the processor configuration from 1 CPU to 2 CPUs. **Step 4:** Open Options → Advanced and check the virtual machine's hardware compatibility settings.

Step 5: If required, use Manage → Change Hardware

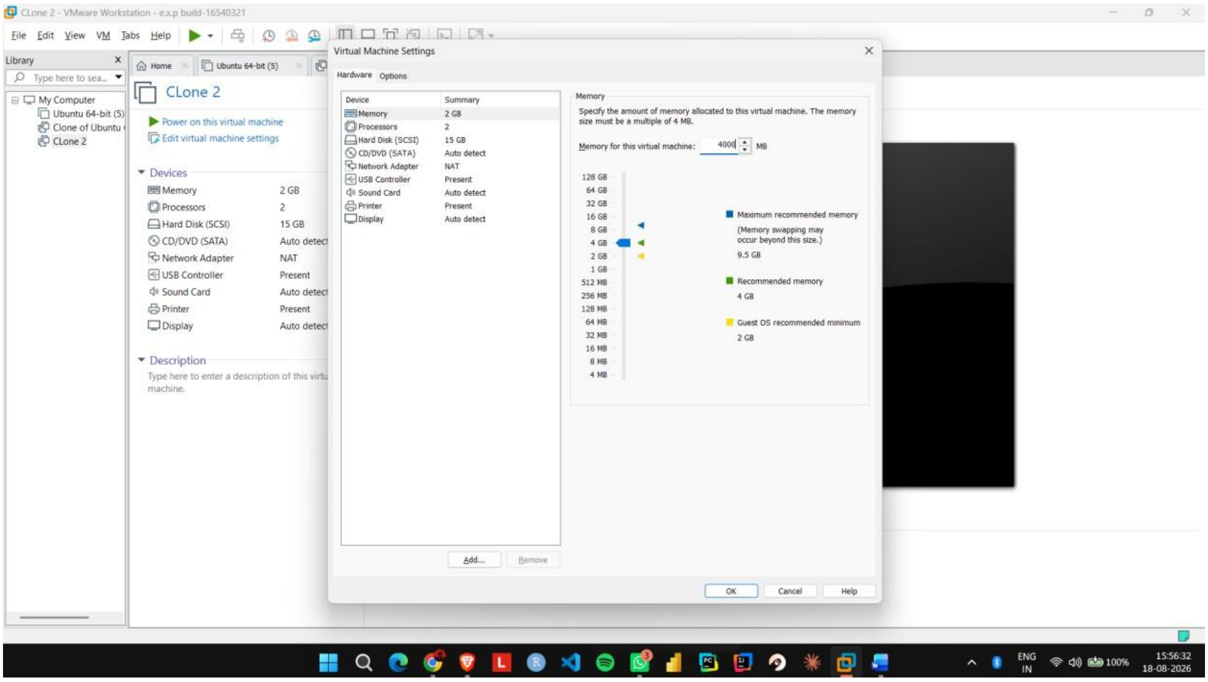
Compatibility and select the required VMware hardware version. **Step 6:** Apply the changes, start Ubuntu, and verify that the modified hardware configuration works correctly.

IMPLEMENTATION:

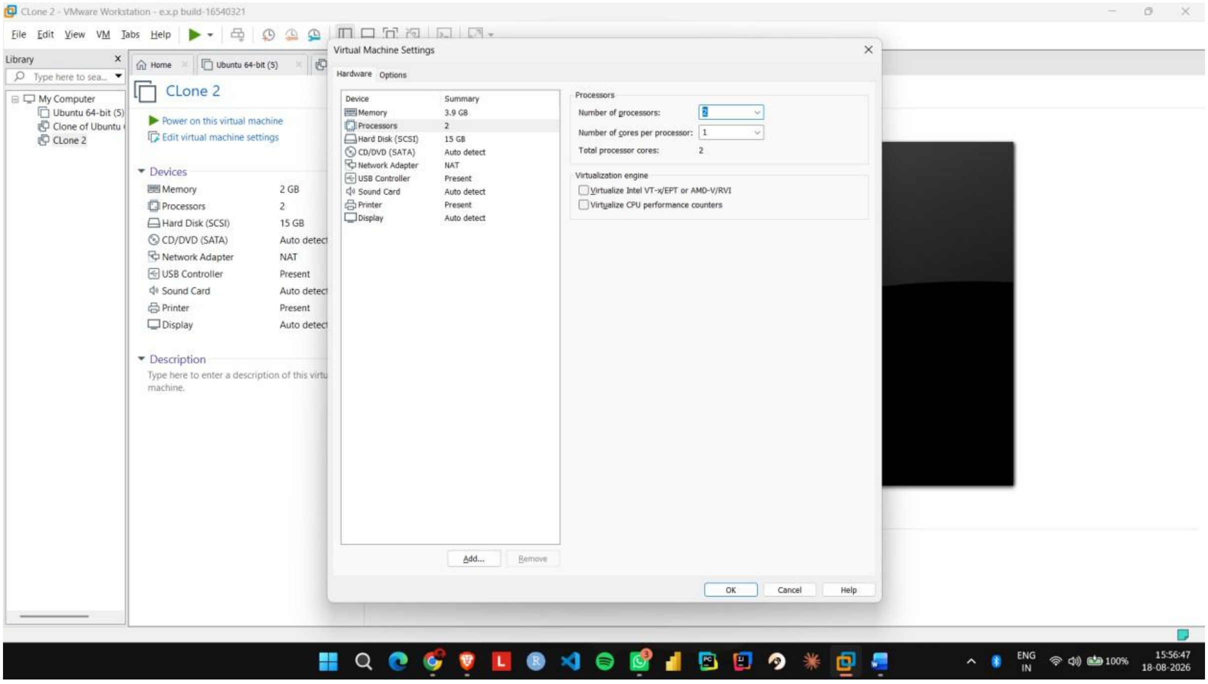
Step 1: Shut down the existing Ubuntu VM and open Edit virtual machine settings.



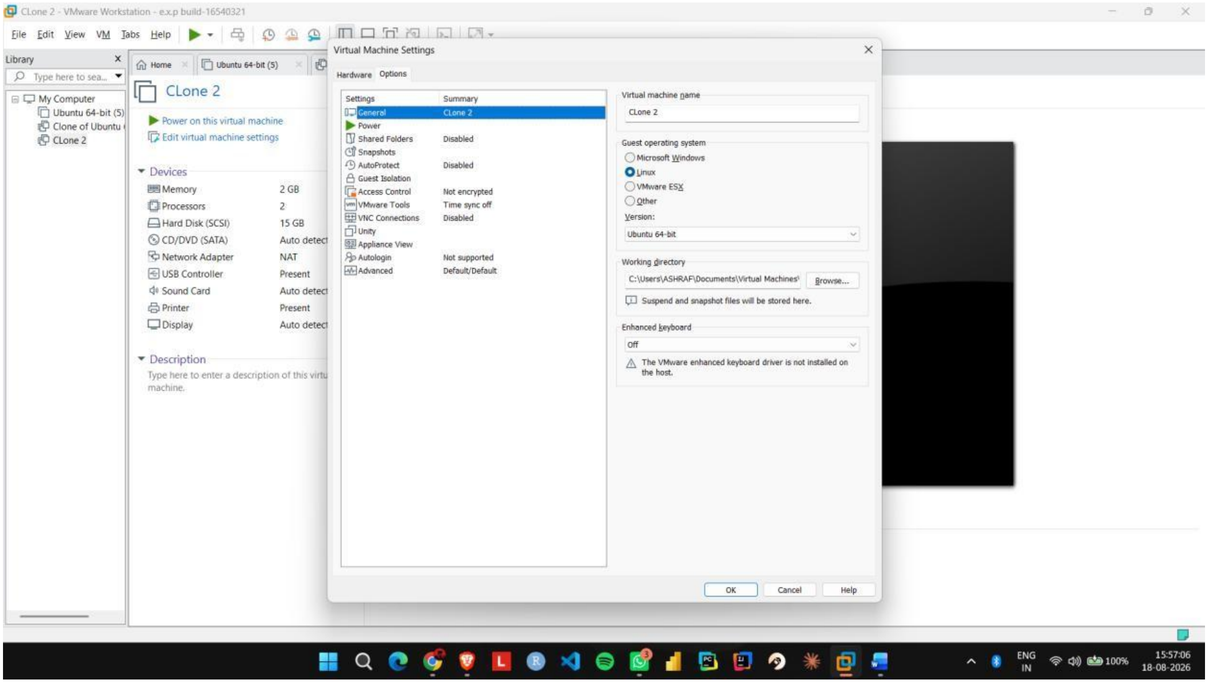
Step 2: Go to Hardware and change the memory from 2 GB to 4 GB.



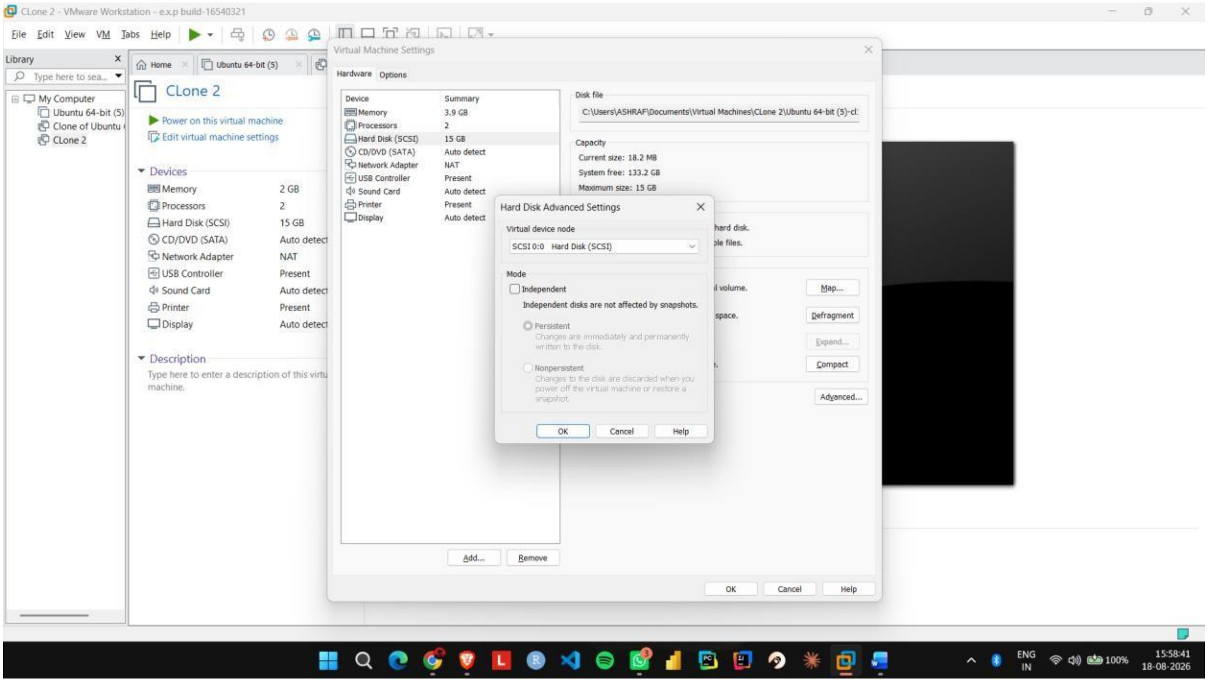
Step 3: Change the processor configuration from 1 CPU to 2 CPUs.



Step 4: Open Options → Advanced and check the virtual machine's hardware compatibility settings.



Step 5: If required, use Manage → Change Hardware Compatibility and select the required VMware hardware version.



Step 6: Apply the changes, start Ubuntu, and verify that the modified hardware configuration works.